Con il contributo di

REGIONE
PIEMONTE

Dipartimento
per le Politiche Giovanili
e il Servizio Civile Universale
Presidenza del Consiglio dei Ministri

PROGETTO FINANZIATO CON IL
FONDO PER LE POLITICHE GIOVANILI

Comune di Vigone

GIOVANI
IN COMUNE!

L’attività e realizzata con il contributo di **Regione Piemonte** e del **Dipartimento per le Politiche giovanili e il Servizio Civile Universale** – per il progetto *“Partecipazione dei giovani alla vita sociale e politica dei territori” – III Edizione”.*

anci
Piemonte

GIOVANI IN COMUNE!

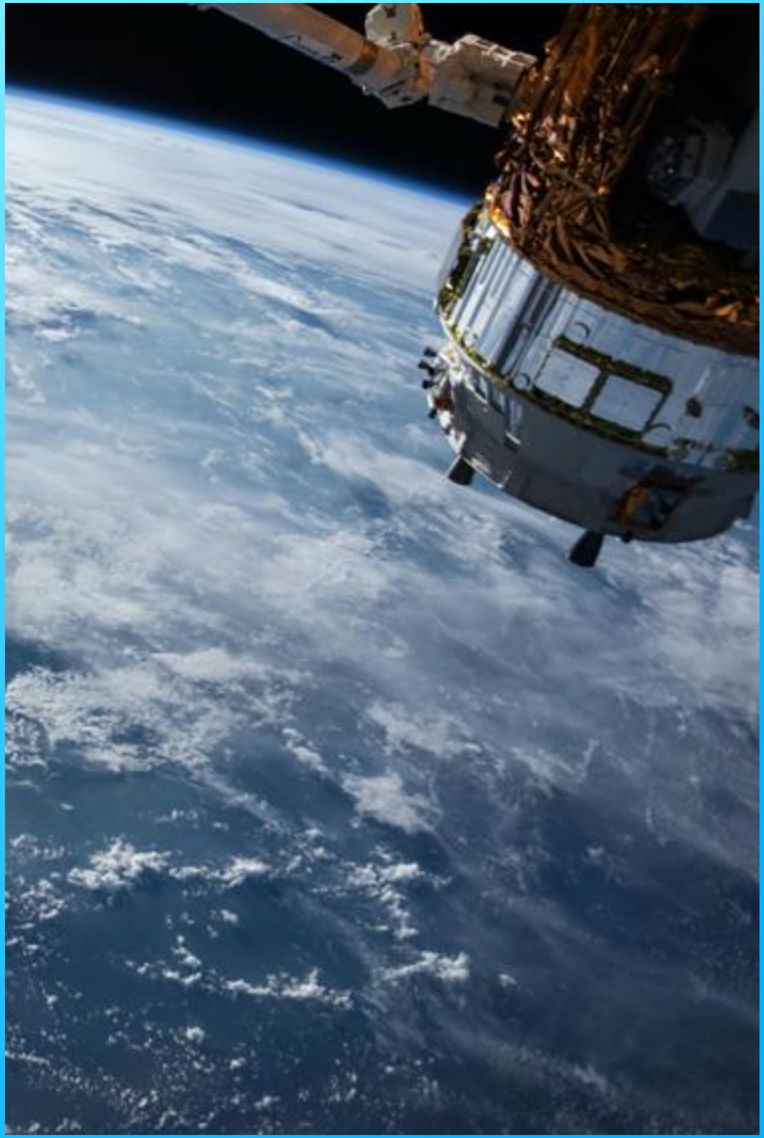
Strumenti digitali per rappresentazione, analisi e gestione del territorio.

Francesca Noardo (Open Geospatial Consortium), Piergiorgio Cipriano (DedaNext)

8 maggio 2025
17:30 – 19:30

Co-funded by the European Union

https://www.ogc.org



What is OGC?

Open Geospatial Consortium

Open Geospatial Consortium

A hub for thought leadership, innovation, and standards for all things related to location

Our Vision

Building the future of location with community and technology for the good of society

Our Mission

Make location information

Findable, Accessible, Interoperable, and Reusable (FAIR)

Our Approach

A proven collaborative and agile process combining consensus-based standards, innovation project, and partnership building

Overview

Innovazione digitale

Focus 1: Dati (geo)spaziali

Focus 2: Casi d’uso e requisiti dei dati

Focus 3: Esempi di casi d’uso e risultati di progetti

- Digitalizzazione dei permessi di costruire
- Strumenti GIS e standard per l’analisi della città e del territorio

DIGITALIZZAZIONE
Uso di tecnologie digitali per cambiare il modello di business e fornire nuove possibilità

Tecnologie

Leggi

Persone

Dati

Processi

Support

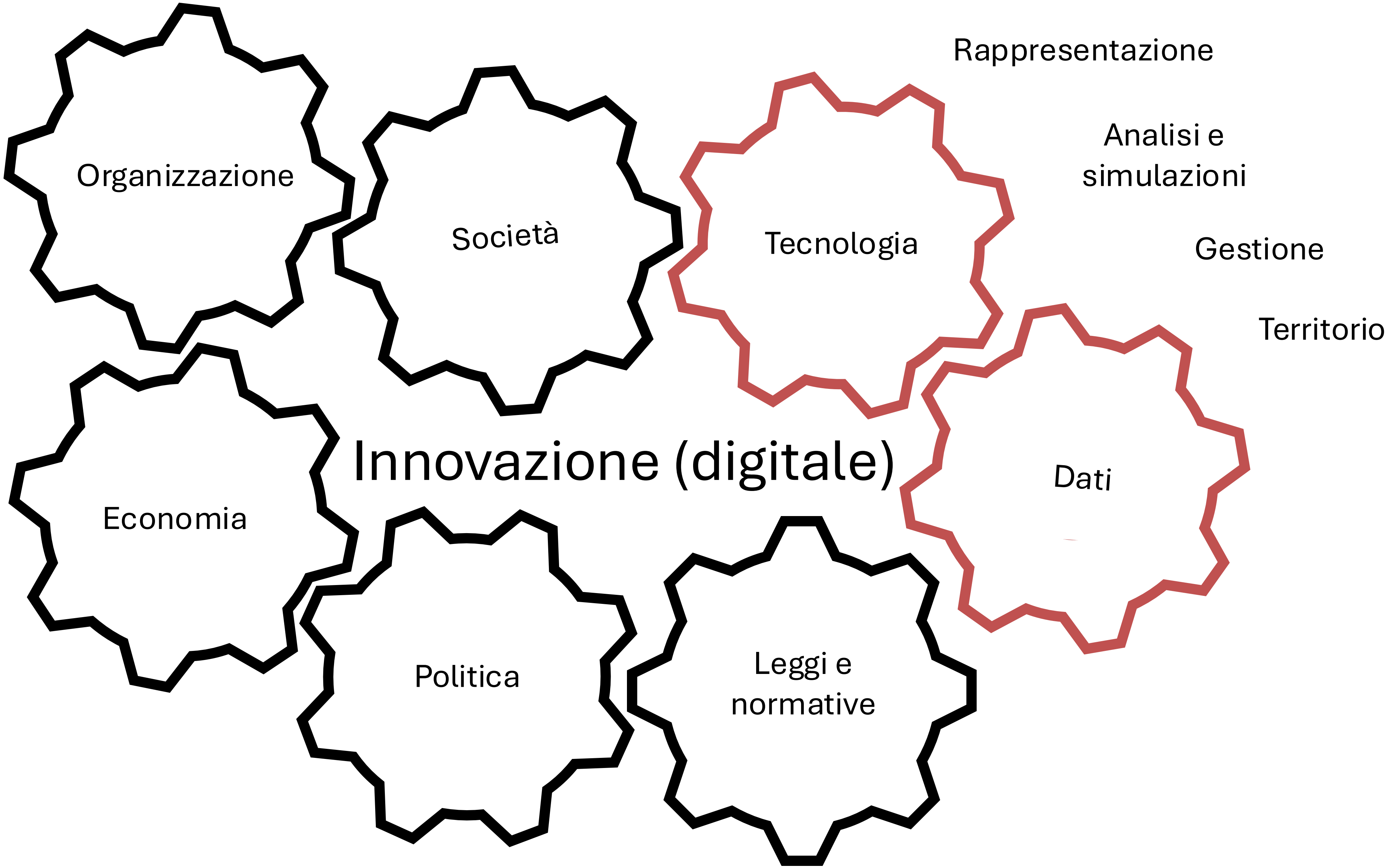
Priority for national and international organizations

- European Commission
- Organisation for Economic Co-operation and Development (OECD)
- G20
- ...

Obiettivi:

- Miglioramento efficienza
- Miglioramento qualità di servizi e città
- Efficace ri-uso di dati
- Abilitazione di analisi automatiche avanzate
- Risparmio tempo e denaro


Organizzazione
Società
Tecnologia
Innovazione (digitale)
Economia
Dati
Politica
Leggi e
normative


Organizzazione
Società
Tecnologia
Economia
Innovazione (digitale)
Dati
Politica
Leggi e normative
Rappresentazione
Analisi e simulazioni
Gestione
Territorio

Innovazione digitale per Rappresentazione, analisi, gestione del territorio

~~Smart city~~

Digital Twins

Data Spaces

Innovazione digitale per Rappresentazione, analisi, gestione del territorio

Digital Twins

MENU Comune di Bologna **iperbole** RETE CIVICA

Home / Tutte le notizie / Bologna avrà un Gemello digitale

Bologna avrà un Gemello digitale

Pubblicato il: **15 settembre 2023**

https://www.comune.bologna.it/notizie/gemello-digitale

Digital Twin | Città | Digitalizzazione | Smart City 6 min

Data Pubblicazione: **01.02.2024**

La nuova era dei Local Digital Twin rivoluziona la gestione delle città europee: da Barcellona a Bologna, niente sarà più come prima

Il testo evidenzia la crescente rivoluzione nel concetto di Local Digital Twin (LDT) in Europa, principalmente attraverso il progetto DUET (Horizon 2020) incentrato sulla City Twin. Questa innovazione incorpora dati dinamici direttamente nel modello LDT, facilitando la pianificazione urbana.

Domenico Santarsiero

https://www.ingenio-web.it/articoli/la-nuova-era-dei-local-digital-twin-rivoluziona-la-gestione-delle-citta-europee-da-barcellona-a-bologna-niente-sara-piu-come-prima/

https://www.ecoincitta.it/smart-city-milano-capoluogo-piu-digitale-deuropa/

Smart city, Milano sarà il capoluogo più digitale d'Europa

BY: MARZIA FIORDALISO / ON: 27 GENNAIO 2023 / IN: MILANO / TAGGED: INQUINAMENTO ATMOSFERICO, RISPARMIO ENERGETICO

Milano avrà il suo **gemello digitale** in 3D che permetterà alla città di funzionare in modo intelligente ed efficiente grazie alla collaborazione che il Comune di Milano porta avanti da diversi anni con Esri Italia; obiettivo promuovere l'innovazione tecnologica per realizzare sistemi a supporto della **smart city**.

Data Spaces

MENU Agenda Digitale Cittadinanza digitale Sicurezza Informatica Sanità digitale

SERVIZI DIGITALI

Le applicazioni del digital twin per i Comuni: l'esperienza di Firenze

Home > Cittadinanza Digitale

Il Digital Twin è un ecosistema di tecnologie che crea repliche digitali precise di oggetti fisici, migliorando la pianificazione urbana. Utilizzato dal Comune di Firenze, ha ottimizzato rilievi e gestione dei dati. Applicato con IA, supporta decisioni su viabilità, sostenibilità e servizi pubblici

Pubblicato il 28 giu 2024

Emanuele Geri
Responsabile EQ "Risorse dati, Open Data e SIT" – Comune di Firenze

https://www.agendadigitale.eu/cittadinanza-digitale/le-applicazioni-del-digital-twin-per-i-comuni-lesperienza-di-firenze/

Menu Cerca **la Repubblica** ABBONATI Accedi

CONTENUTO PER GLI ABBONATI PREMIUM

Digital twin, droni e IA, il "gemello virtuale" che aiuta la città a diventare efficiente

di Stefania Aoi

Il progetto di Comune e Politecnico permetterà di prendere decisioni per esempio nell'ambito del verde pubblico o dei cantieri stradali ed evitare sprechi di risorse

19 APRILE 2024 ALLE 09:12 2 MINUTI DI LETTURA

https://torino.repubblica.it/cronaca/2024/04/19/news/digital_twintorino_politecnico_comune_efficienza-422602640/

Innovazione digitale per Rappresentazione, analisi, gestione del territorio

Digital Twins

https://www.smartbuildingitalia.it/news/smart-city/digital-twin-smart-city/

Data Spaces

https://www.linkedin.com/pulse/top-use-cases-digital-twins-c3advisory/

https://www.ingenio-web.it/articoli/digital-twin-o-gemello-digitale-7-modi-in-cui-le-citta-possono-trarne-vantaggio-dal-loro-uso/

Open Geospatial Consortium.

deda.next envision public services

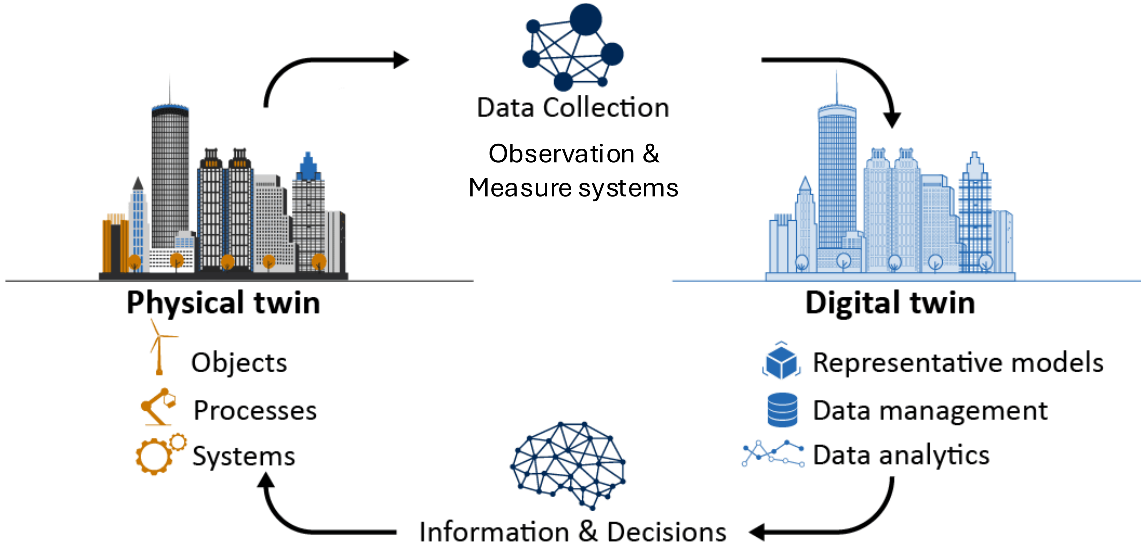
Digital Twins

L'obiettivo dei gemelli digitali su scala urbana è la **trasformazione** del modo in cui le città vengono pianificate, costruite e gestite al fine di fornire **servizi migliori** e rendere l'ambiente urbano più vivibile, inclusivo, sicuro, resiliente e sostenibile.

Il "**gemellaggio**" è stato utilizzato dalla NASA negli anni '60 con la pratica di costruire sistemi fisicamente duplicati a livello del suolo per adattarli al sistema nello spazio.
Il concetto di **gemello digitale** (Digital Twin) è nato nel 2002 nell'ambito dell'ingegneria industriale e meccanica. Un gemello digitale è, in sintesi, una rappresentazione digitale del mondo reale, che **rispecchia continuamente**.

City information modelling and urban digital twins IEC Technology report

Data Spaces



- Model
- Spatial dimension (3D)
- Exact mirror concept → fit for purpose
- Real-time and dynamic information
- Analysis, simulations and predictions
- Automatic interaction with reality

https://www.gao.gov/products/gao-23-106453

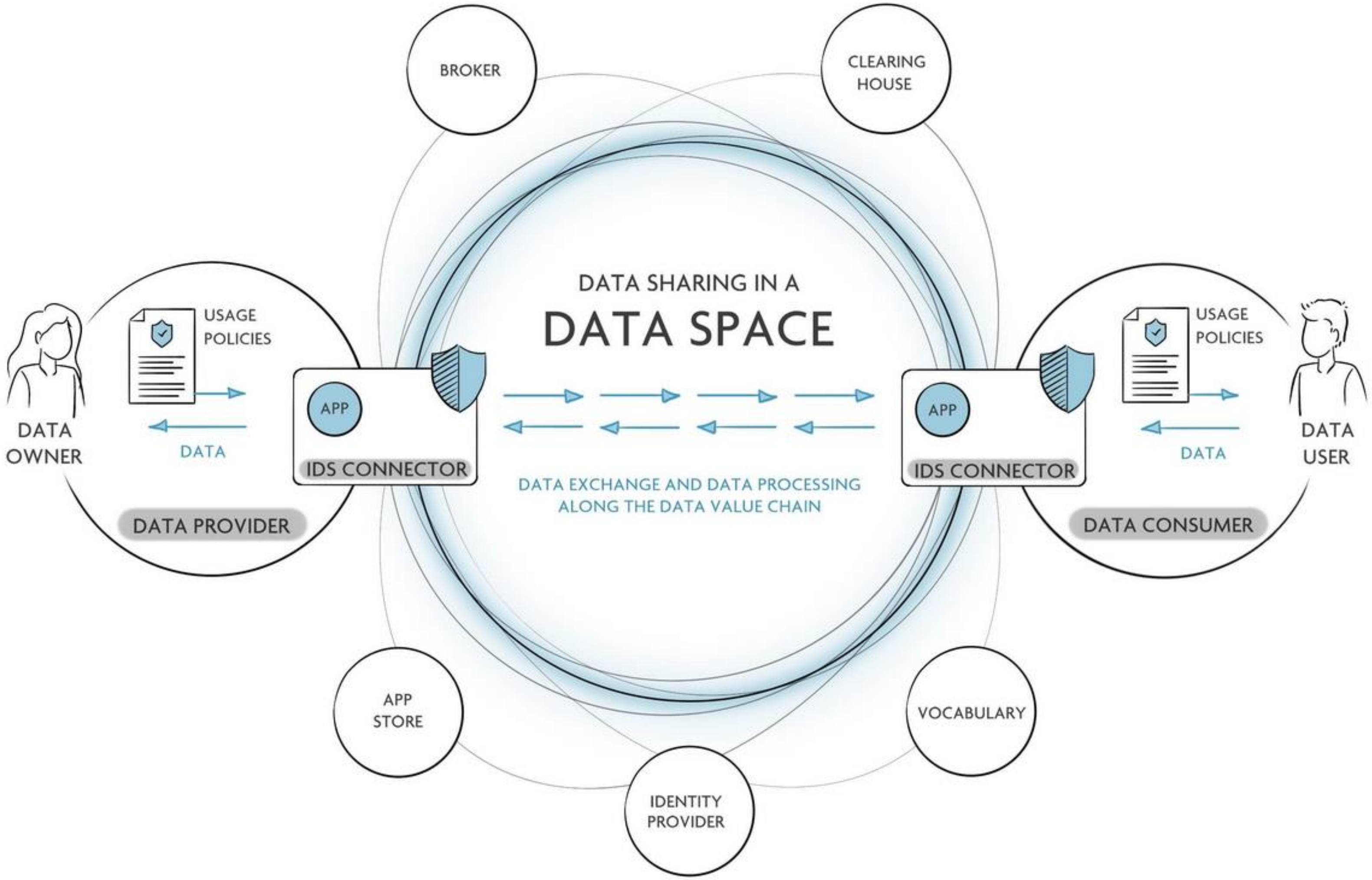
Nativi, S.; Mazzetti, P.; Craglia, M. Digital Ecosystems for Developing Digital Twins of the Earth: The Destination Earth Case. Remote Sens. 2021, 13, 2119. https://doi.org/10.3390/rs13112119

A **distributed system** defined by a governance framework that enables secure and trustworthy **data transactions** between participants while supporting trust and data sovereignty. A data space is implemented by **one or more infrastructures** and enables **one or more use cases**.
(Data Space Support Center, Glossary - https://dssc.eu/)

Digital Twins

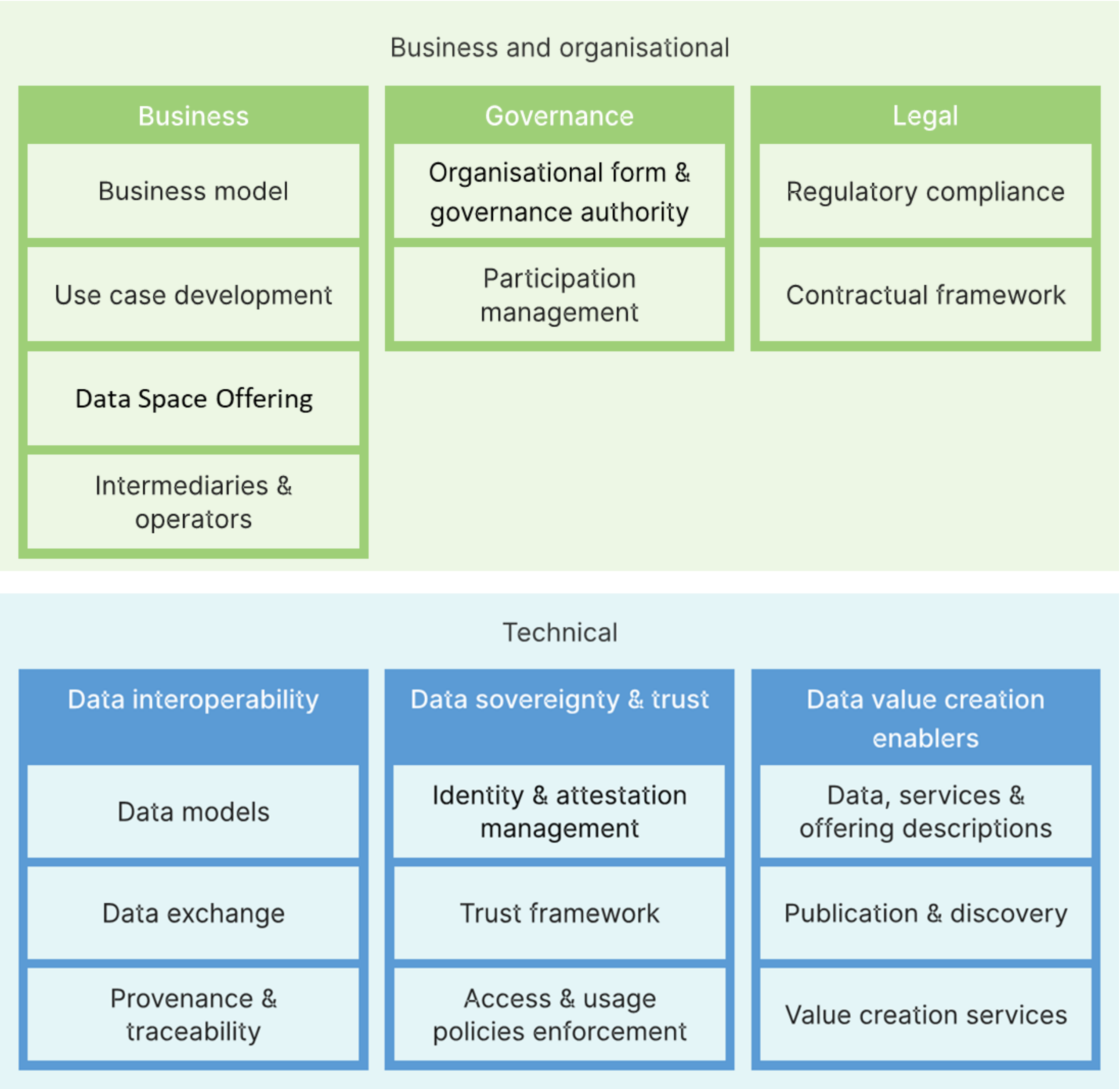
Data Spaces

Una **infrastruttura decentralizzata** per la raccolta, l'elaborazione e lo scambio di dati affidabili basata su principi comunemente concordati, ha l'obiettivo di facilitare la combinazione di **dati eterogenei** provenienti da **fonti diverse**.
(Comune di Ferrara - progetto USAGE - Atto Costitutivo Data Space Urbano)



Digital Twins

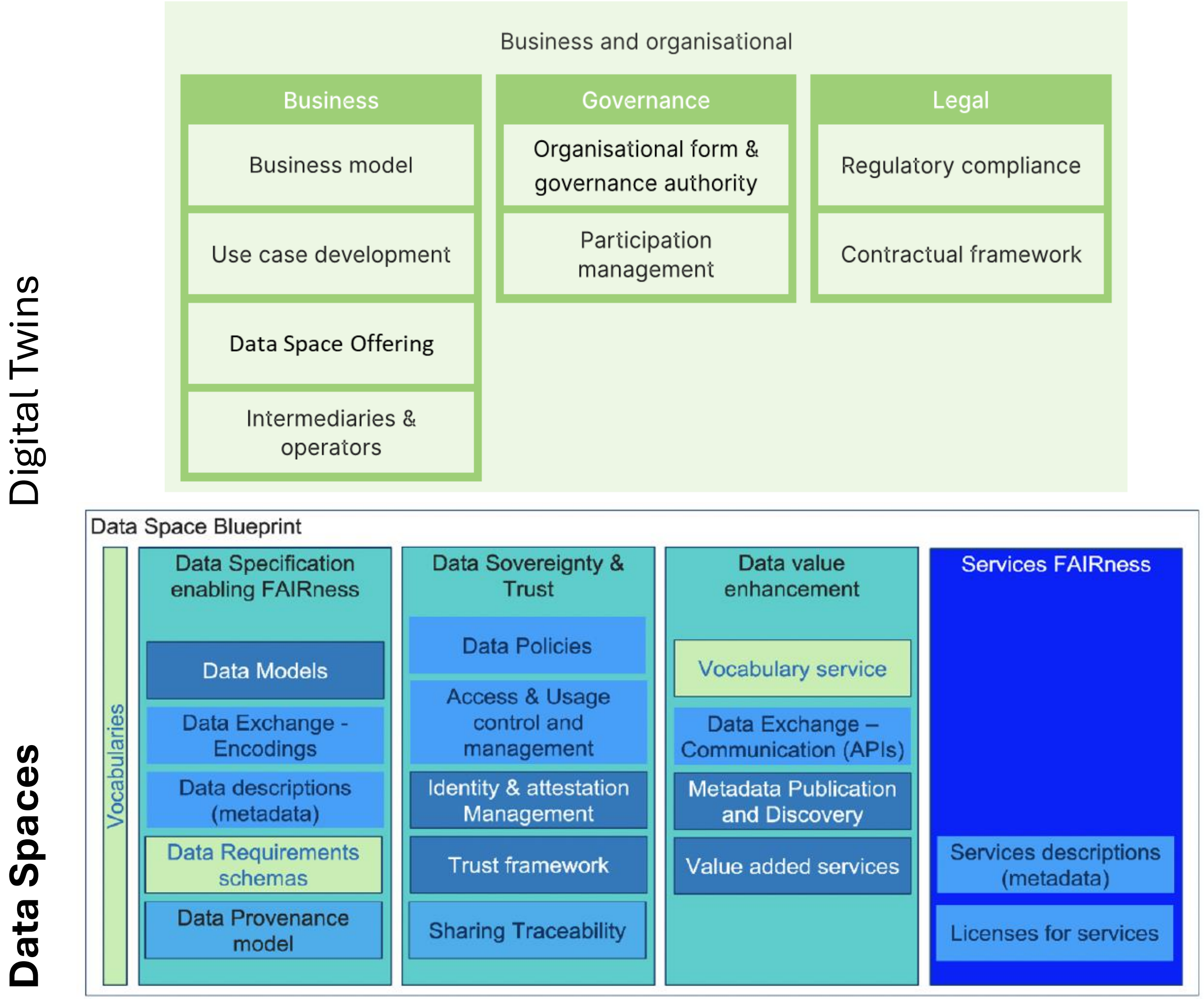
Data Spaces



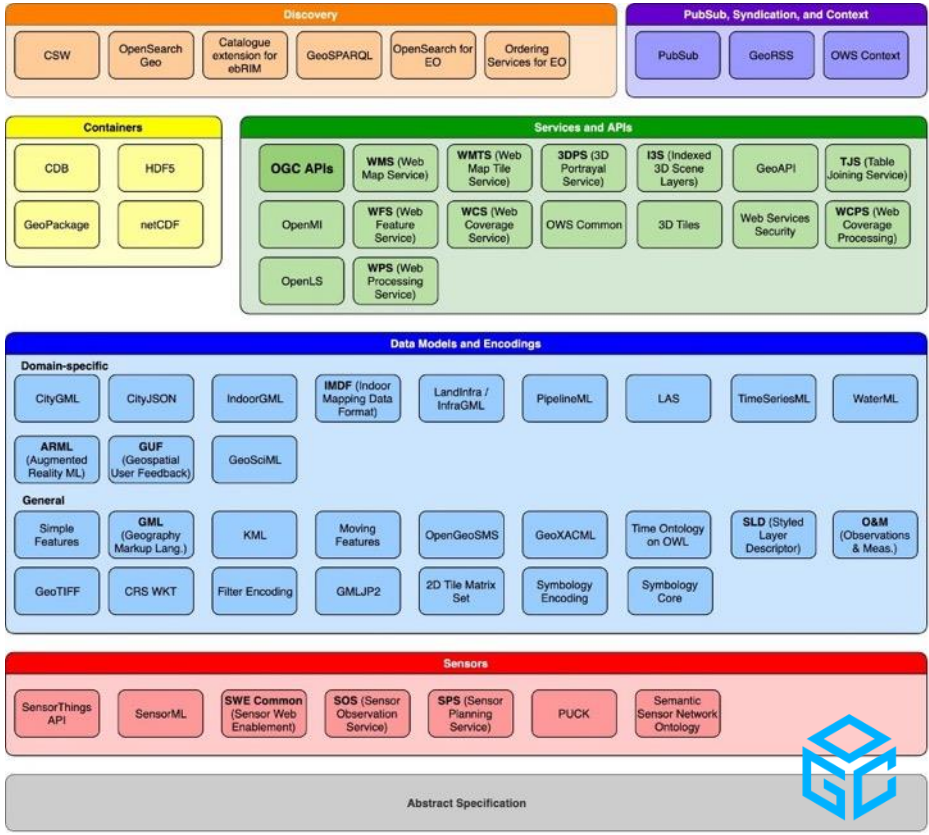
DSSC, 2025. Blueprint 2.0
https://dssc.eu/space/BVE2/1071251457/Data+Spaces+Blueprint+v2.0+-+Home

Strumenti digitali per rappresentazione, analisi e gestione del territorio.

Francesca Noardo (OGC), Piergiorgio Cipriano (DedaNext)



https://www.mdpi.com/2072-4292/16/20/3824#table_body_display_remotesensing-16-03824-t003



https://www.ogc.org/standards/

And more …

Digital Twins

Digital Twins

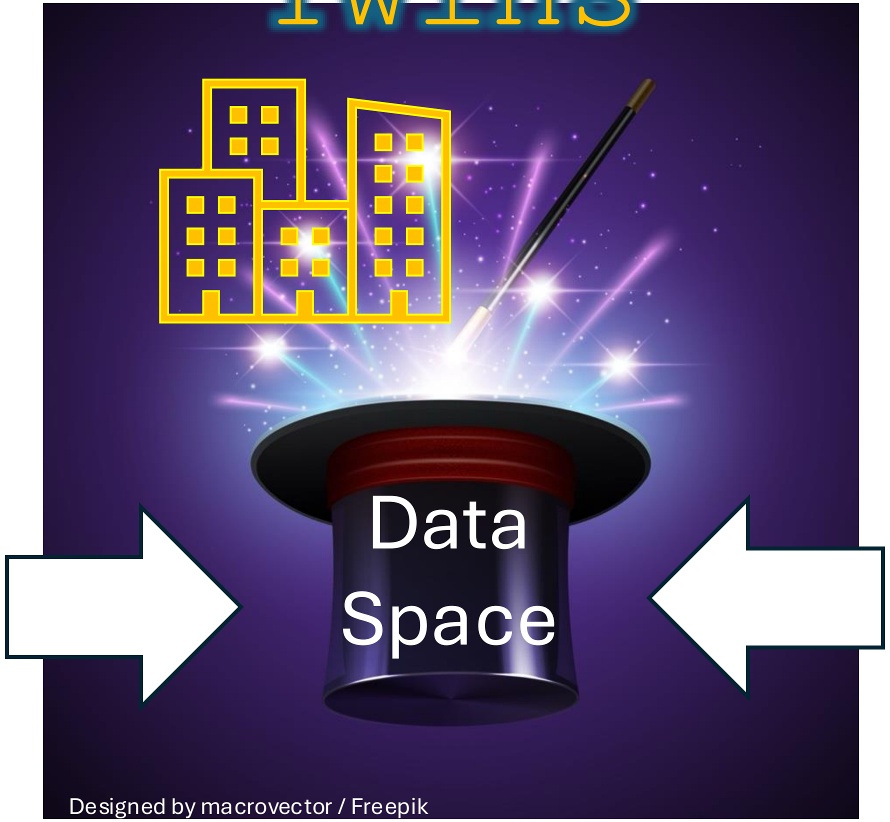
Multiple functions

Real-time data

Data Spaces

Data integration

Multiple accesses



Designed by macrovector / Freepik

Proprietary data

Distributed data

Legal restrictions

…

Digital Twins and Data Spaces use cases

Automazione processi:

- Permessi di costruire digitali
- …

Analisi e applicazioni a supporto di:

- Sostenibilità ambientale ed energetica
- Gestione efficiente città e infrastrutture
- Inclusività e aspetti sociali
- Beni culturali
- Turismo
- …

Focus1: Dati e sistemi informativi spaziali

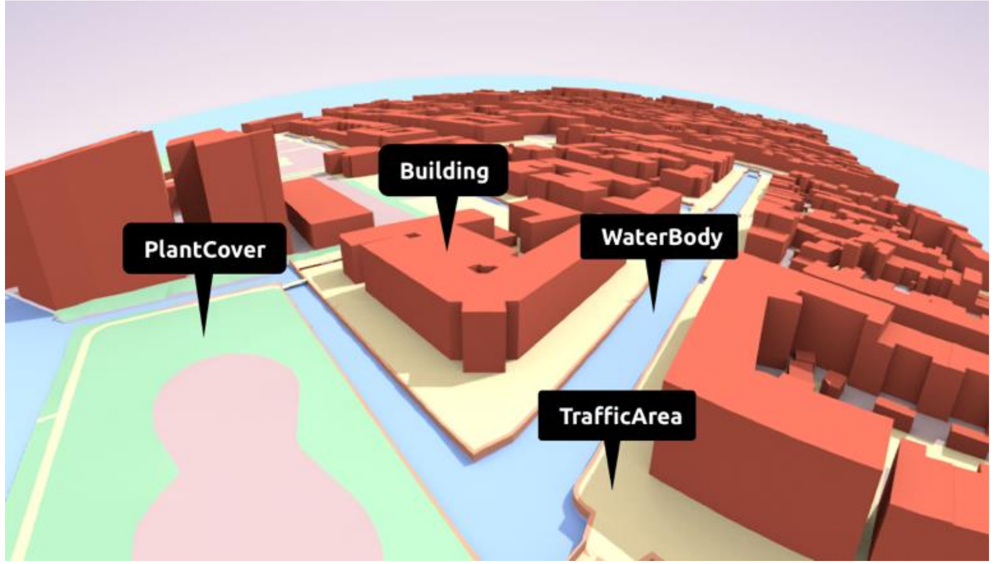
Focus2: Definizione di casi d'uso e requisiti

Focus3: Risultati di progetti per:
- Permessi di costruire digitali
- Sostenibilità ambientale (Green Deal)

Focus1: Dati e sistemi informativi spaziali

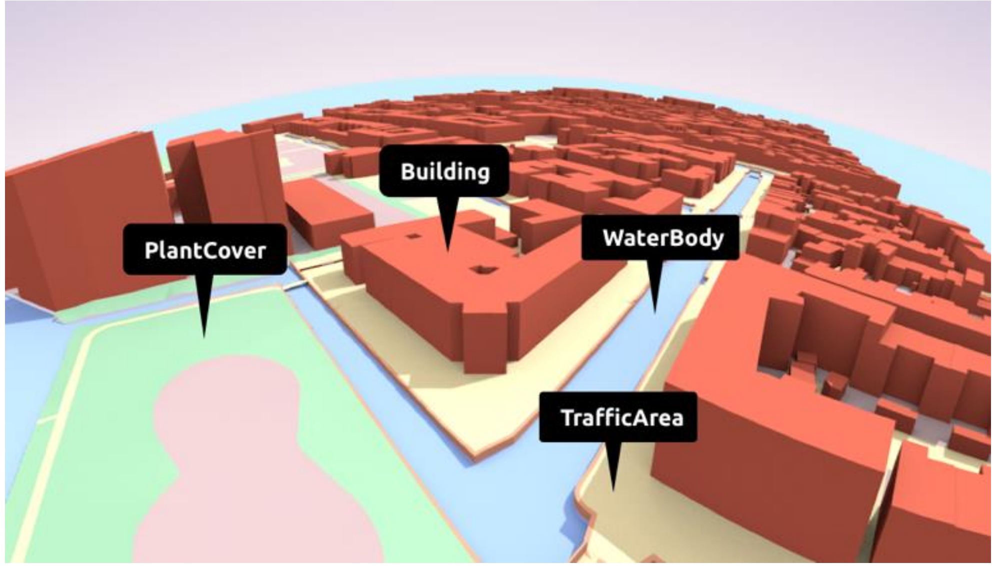
Geoinformation
Geographical Information Systems
3D city model

Building **I**nformation **M**odel

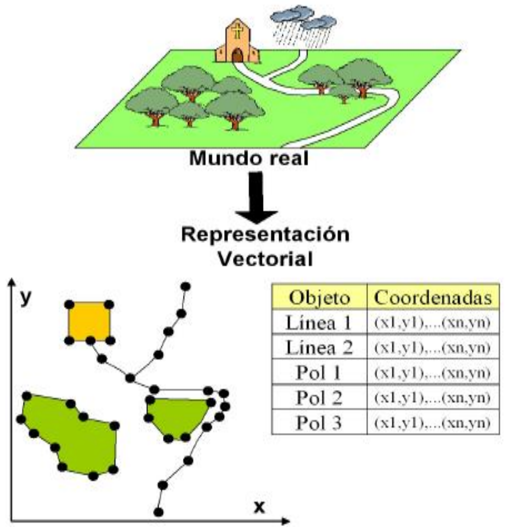


Focus1: Dati e sistemi informativi spaziali

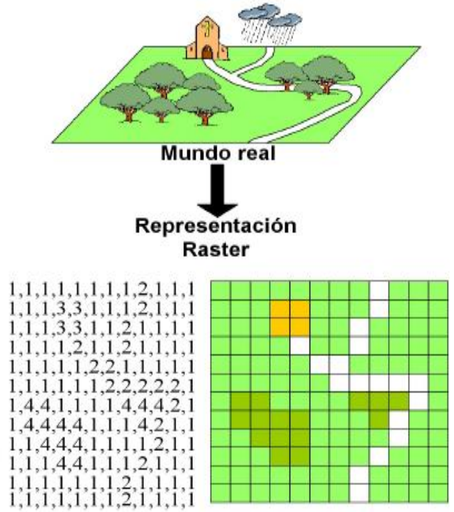
Geoinformation
Geographical Information Systems
3D city model

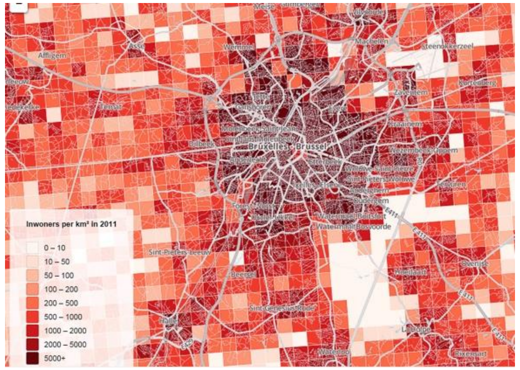


Geodata - Data with location

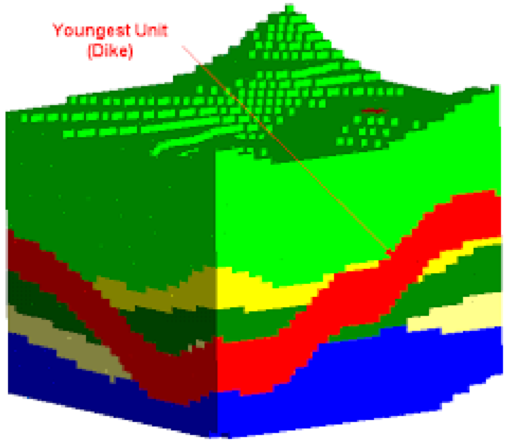


Vector (objects)

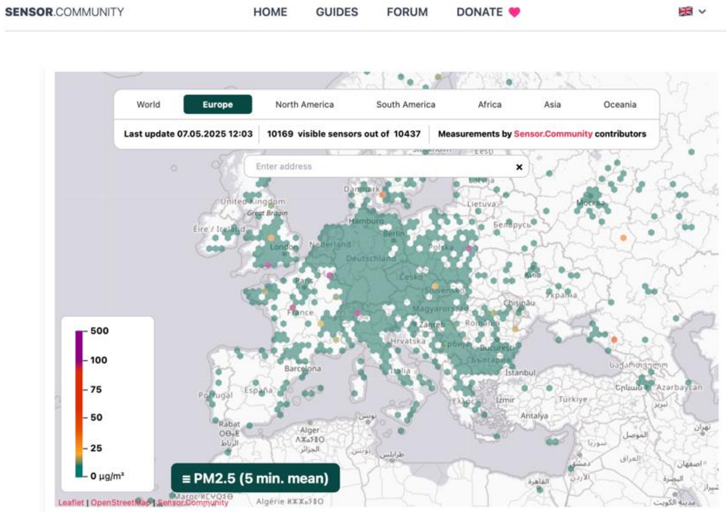




Raster



Voxel



Sensor

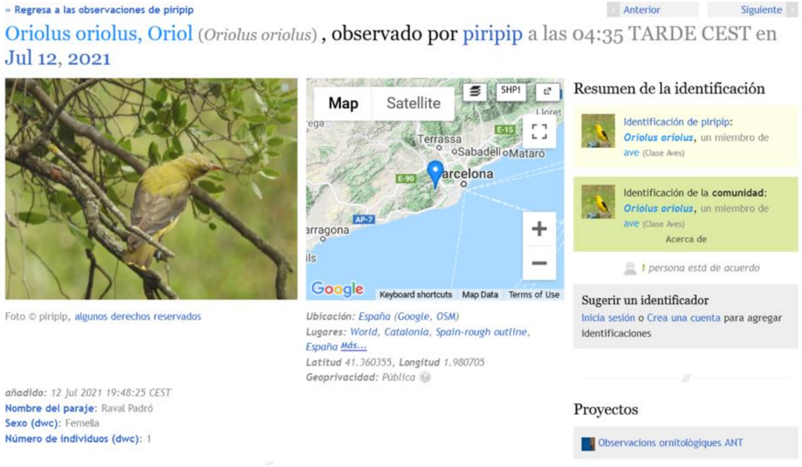


Figure 7: A screenshot of the web interface in Natusfera portal showing observation group ID 313411 (source: https://natusfera.gbif.es/observations/313411)

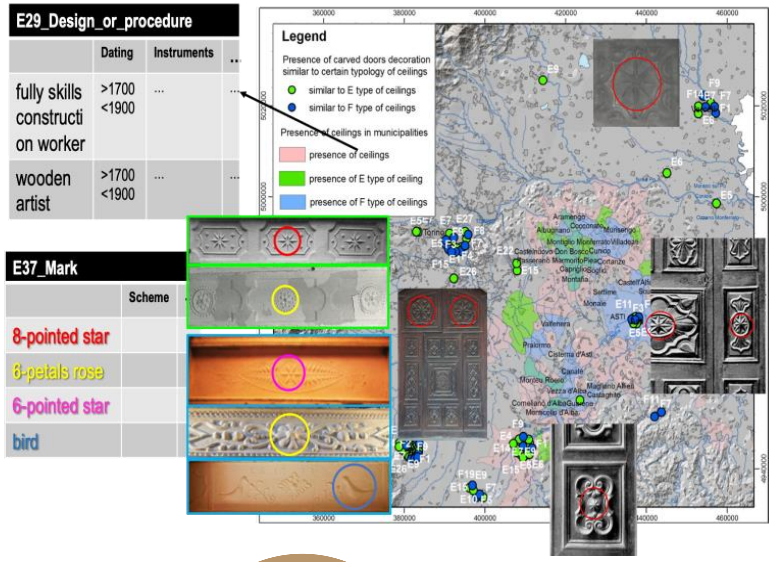
Geographical Information System (GIS) – Sistema Informativo Territoriale (SIT)

It's an association of people, machines, data and procedures working together to collect, manage and distribute information.

Data

Data base

People



Protocols/ procedures

Software

Hardware

Networks

Including **spatial** data

→ Geographical Information Systems

GIS

(digital maps)

Computer-based information system that enables *capture*, *storage*, *retrieval*, *modelling*, *sharing*, *manipulation*, *analysis and presentation* of **geographically referenced data**.

(Worboys, Duckham, 2004)


Open Geospatial Consortium
REAL WORLD
GIS WORLD
Building Data
Vegetation Data
Streets Data
Pipelines Data
Orthophoto

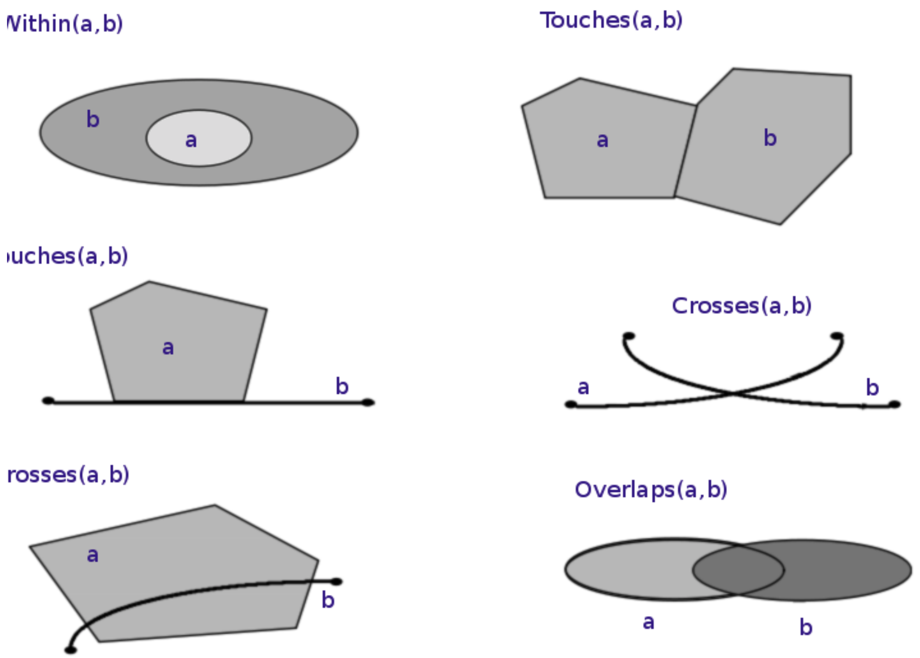
Vector Spatial Analysis

Query spaziali
Capacità di effettuare operazioni basate sulle caratteristiche spaziali dei dati: la loro geometria.

Seleziona per posizione
Seleziona le caratteristiche in base alla loro posizione e relazione spaziale.

Ad esempio:

- *Quali sono gli edifici a meno di 200 m da una strada?*
- *Quali sono le fermate dell'autobus su una strada?*
- *Quali sono gli edifici che confinano con un giardino?*
- *Seleziona gli edifici all'interno di un quartiere,*
- *Punti di utilizzo del suolo all'interno di un edificio,*
- *Strade che attraversano un'area, ecc.*



https://www.qgistutorials.com/en/docs/performing_spatial_queries.html
http://desktop.arcgis.com/en/arcmap/latest/map/working-with-layers/using-select-by-location.htm

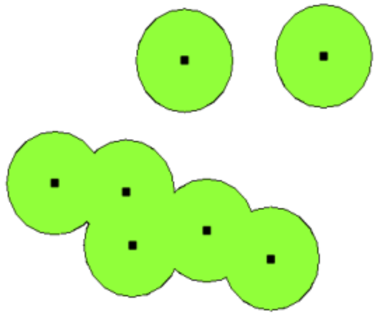
Vector Spatial Analysis

Buffering genera un'area (un poligono vettoriale) entro una distanza specificata dagli oggetti (vettoriali) selezionati (la zona cuscinetto).

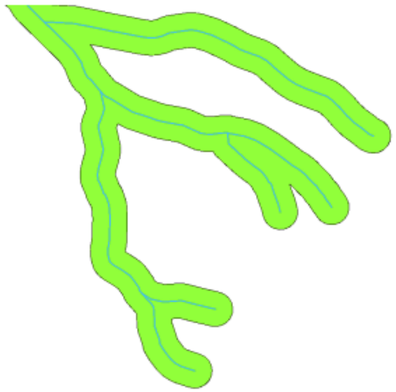
Esempi di applicazione:

- *protezione ambientale*
- *protezione di zone residenziali e commerciali da incidenti industriali, inquinamento e rumore (cinture verdi e zone cuscinetto attorno a industrie, grandi impianti, ferrovie, autostrade...)*
- *o disastri naturali (aree alluvionali attorno a fiumi, laghi, coste)*
- *zone di confine tra paesi*
- *Rispetto delle aree circostanti il patrimonio storico*
- *Creazione di zone cuscinetto attorno a linee di servizio o radici di alberi nel terreno*
- ...

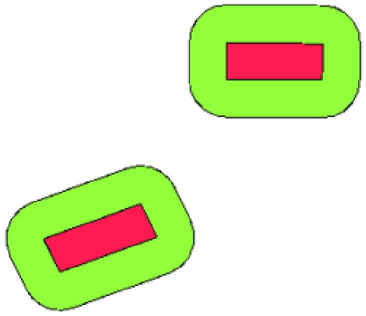
https://docs.qgis.org/latest/en/docs/gentle_gis_introduction/vector_spatial_analysis_buffers.html
http://desktop.arcgis.com/en/arcmap/latest/analyze/commonly-used-tools/proximity-analysis.htm



A buffer zone around vector points.



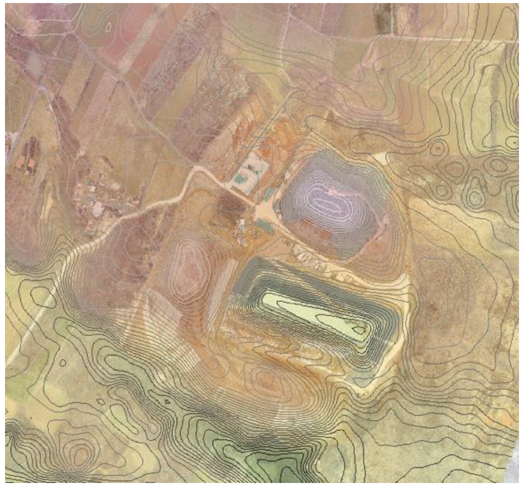
A buffer zone around vector polylines.

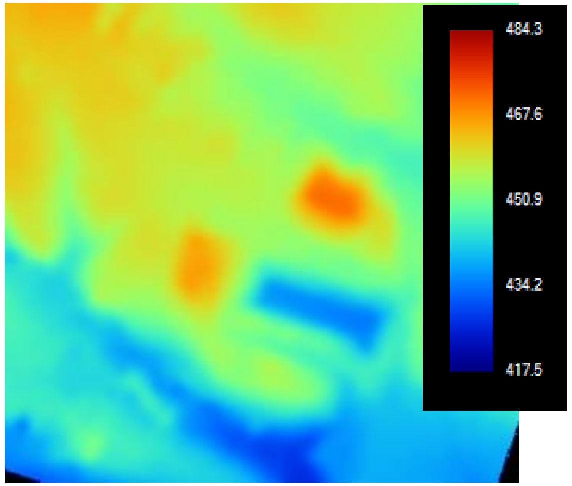


A buffer zone around vector polylines.

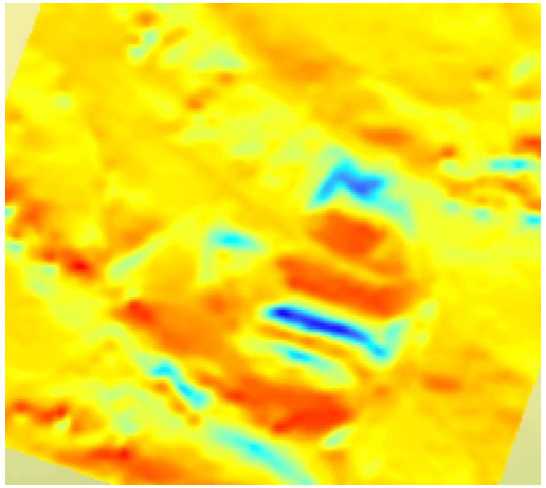
Raster Spatial Analysis: Surface analysis

Contours & DTM (Elevation values)



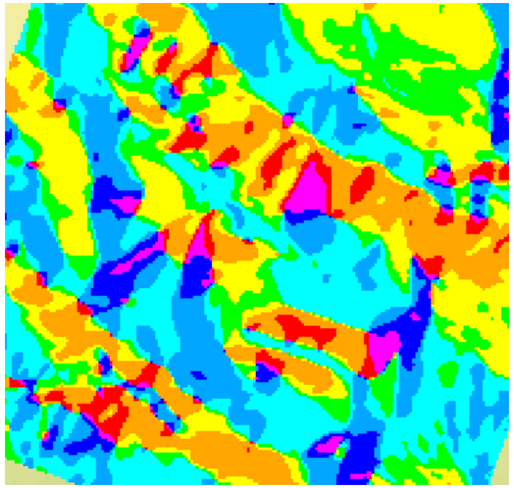


Solar radiation values



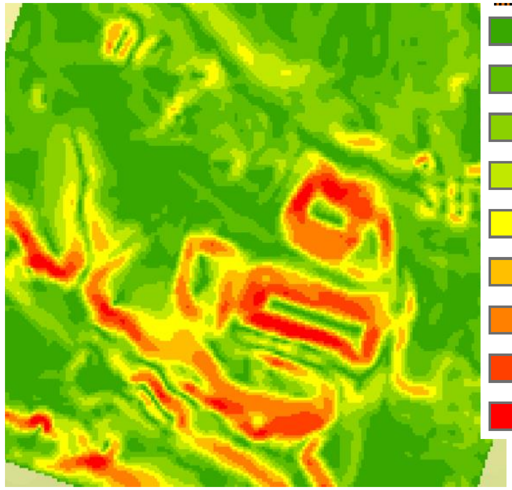
Value
High : 555213
Low : 348792

Aspect values (direction of surfaces)



- Flat (-1)
- North (0-22.5)
- Northeast (22.5-67.5)
- East (67.5-112.5)
- Southeast (112.5-157.5)
- South (157.5-202.5)
- Southwest (202.5-247.5)
- West (247.5-292.5)
- Northwest (292.5-337.5)
- North (337.5-360)

Slope values

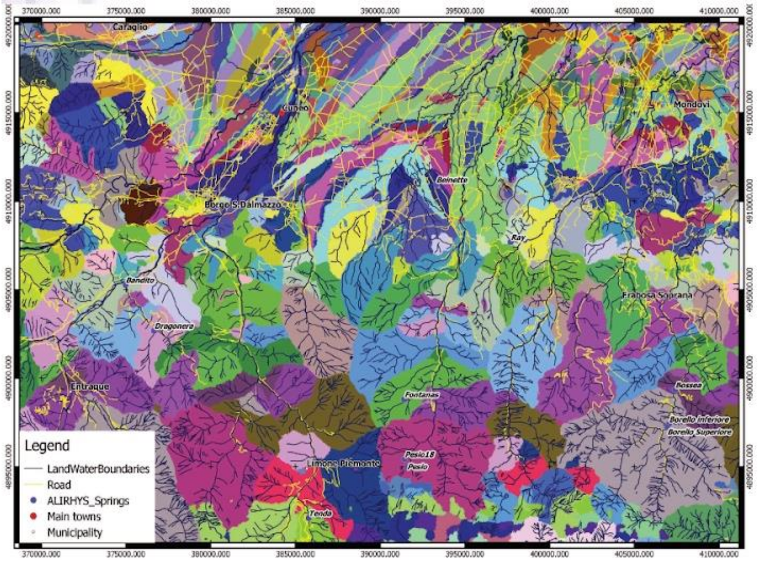


- 0,015912568 - 1,952391311
- 1,952391312 - 3,404750368
- 3,404750369 - 5,244405174
- 5,244405175 - 7,374531791
- 7,374531792 - 9,79513022
- 9,795130221 - 12,31255259
- 12,3125526 - 15,12044676
- 15,12044677 - 18,6061085
- 18,60610851 - 24,70601654

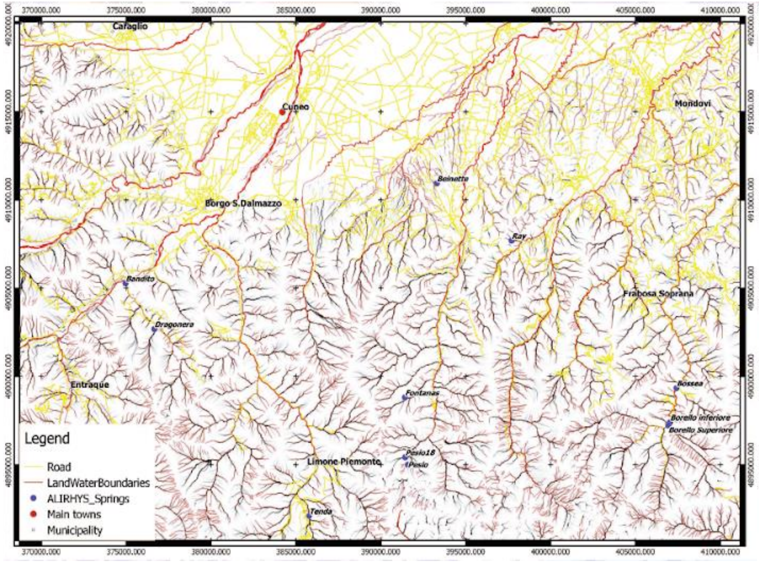
Raster Spatial Analysis: Surface analysis

Hydrological and morphological studies

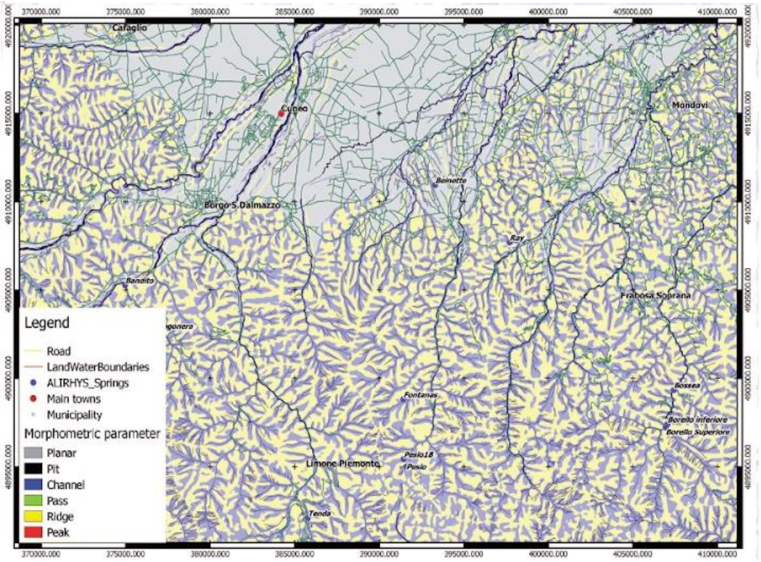
Sink watershed



Catchment computations



Morphometric parameters



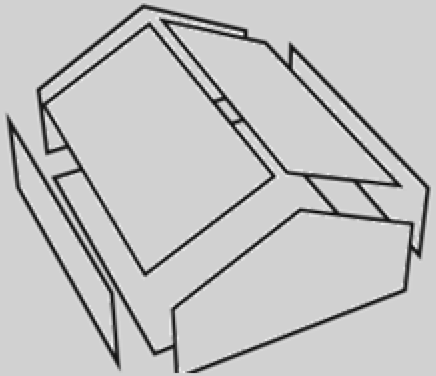
3D geoinformation
3D city models

Evolution of **GIS**;

In **cartographic** field and **city management** (not pushed by industry!);

Medium levels of detail objects (e.g. ~**20 cm** accuracy);

Geometry

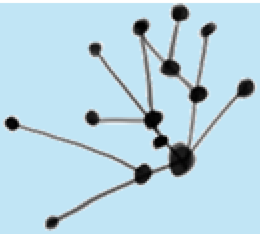


Boundary representation

Storage of all the coordinates composing a surface (explicit representation)

Georeferencing

Semantics



City representation and management aims

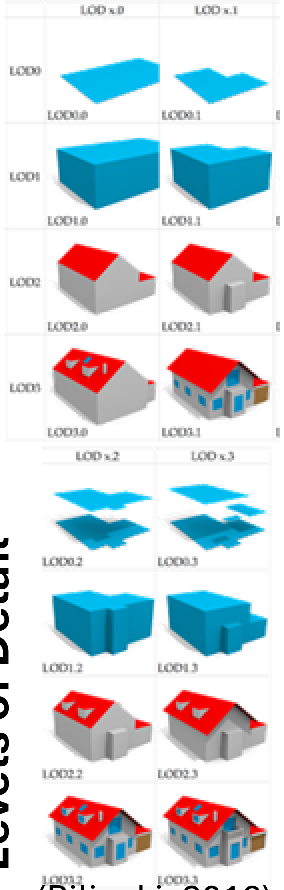
City

Buildings

Vegetation

Water

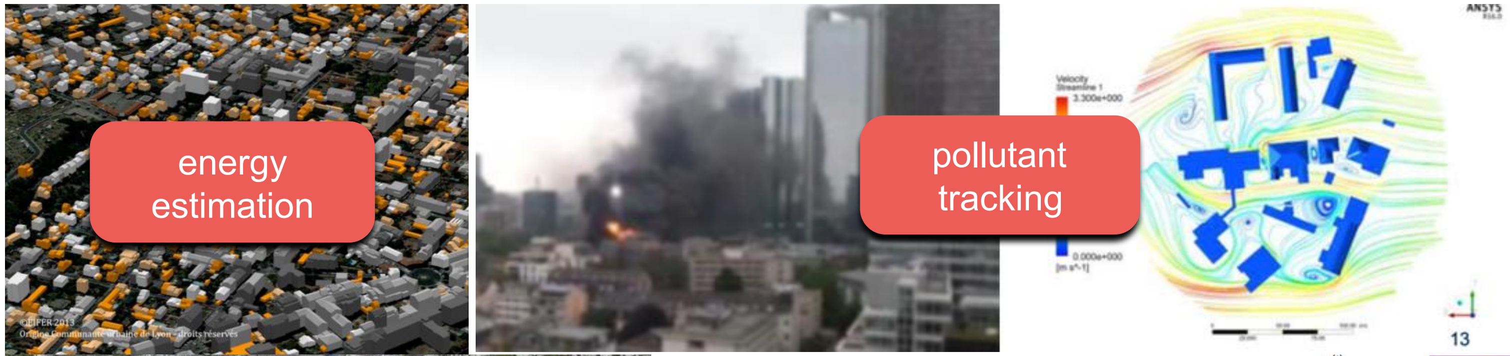
Roads



Levels of Detail

(Biljecki, 2016)

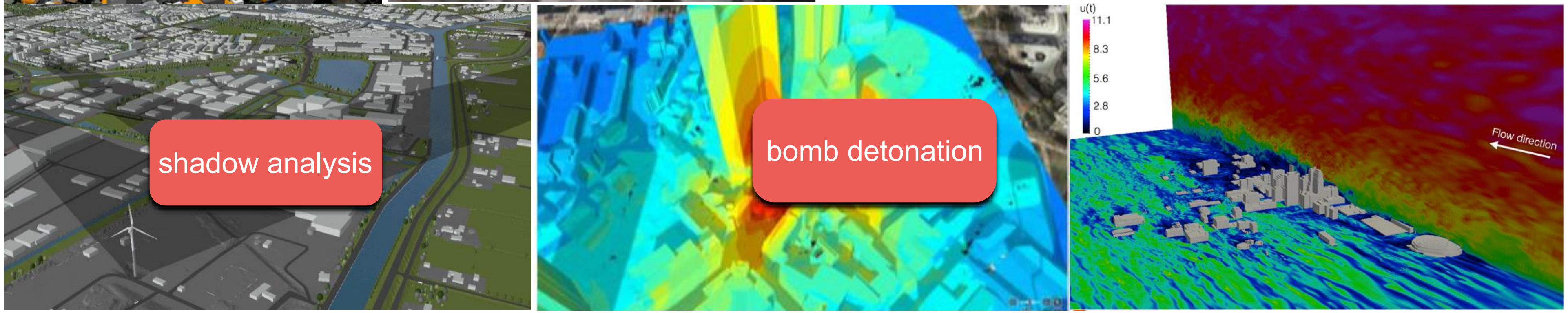
3D geoinformation - 3D city models use cases

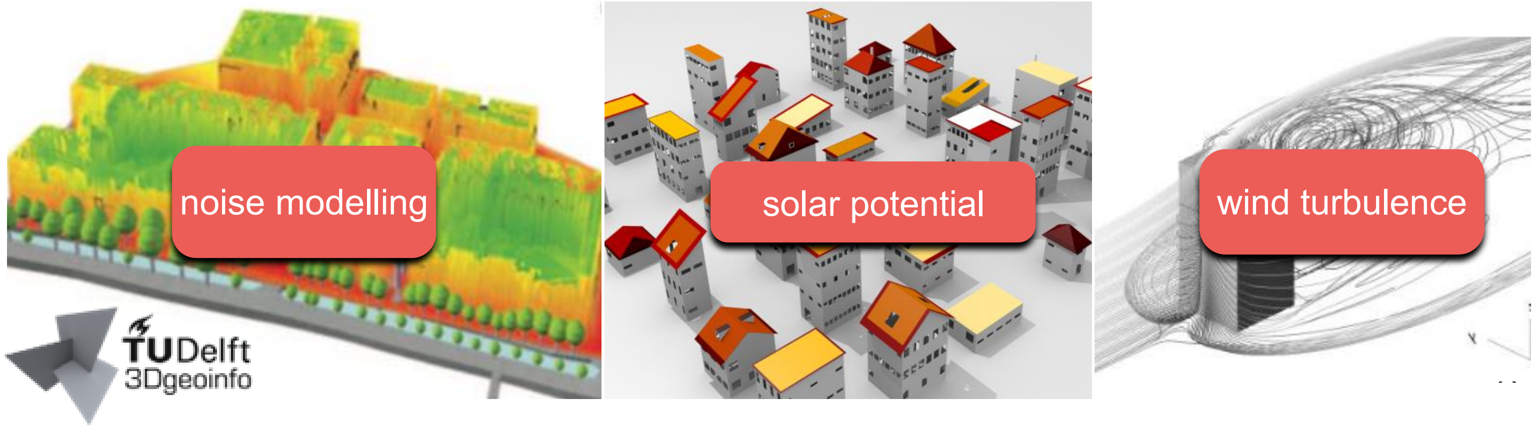


Navigation

Flood simulations

Multivariate analysis





3D cadastre

3D archive

Multitemporal analysis

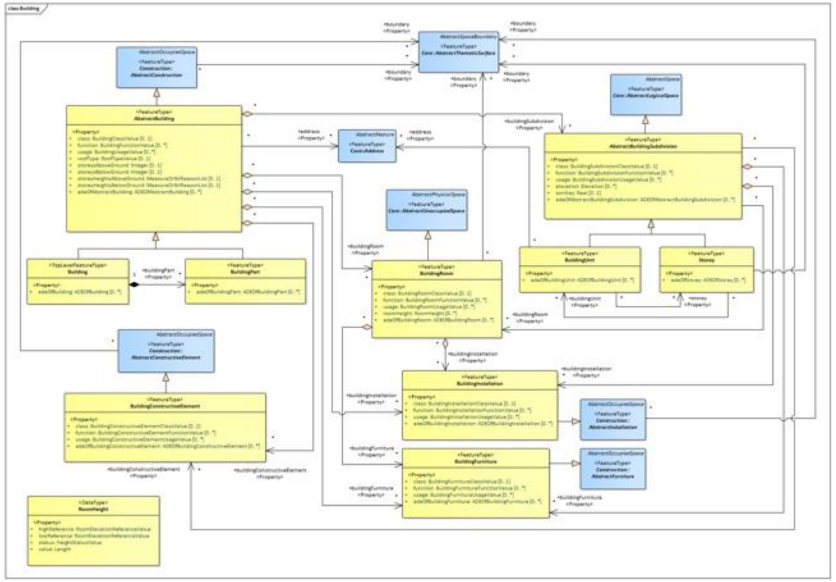
Risk assessment…

OGC CityGML & CityJSON – open standard for 3D city models

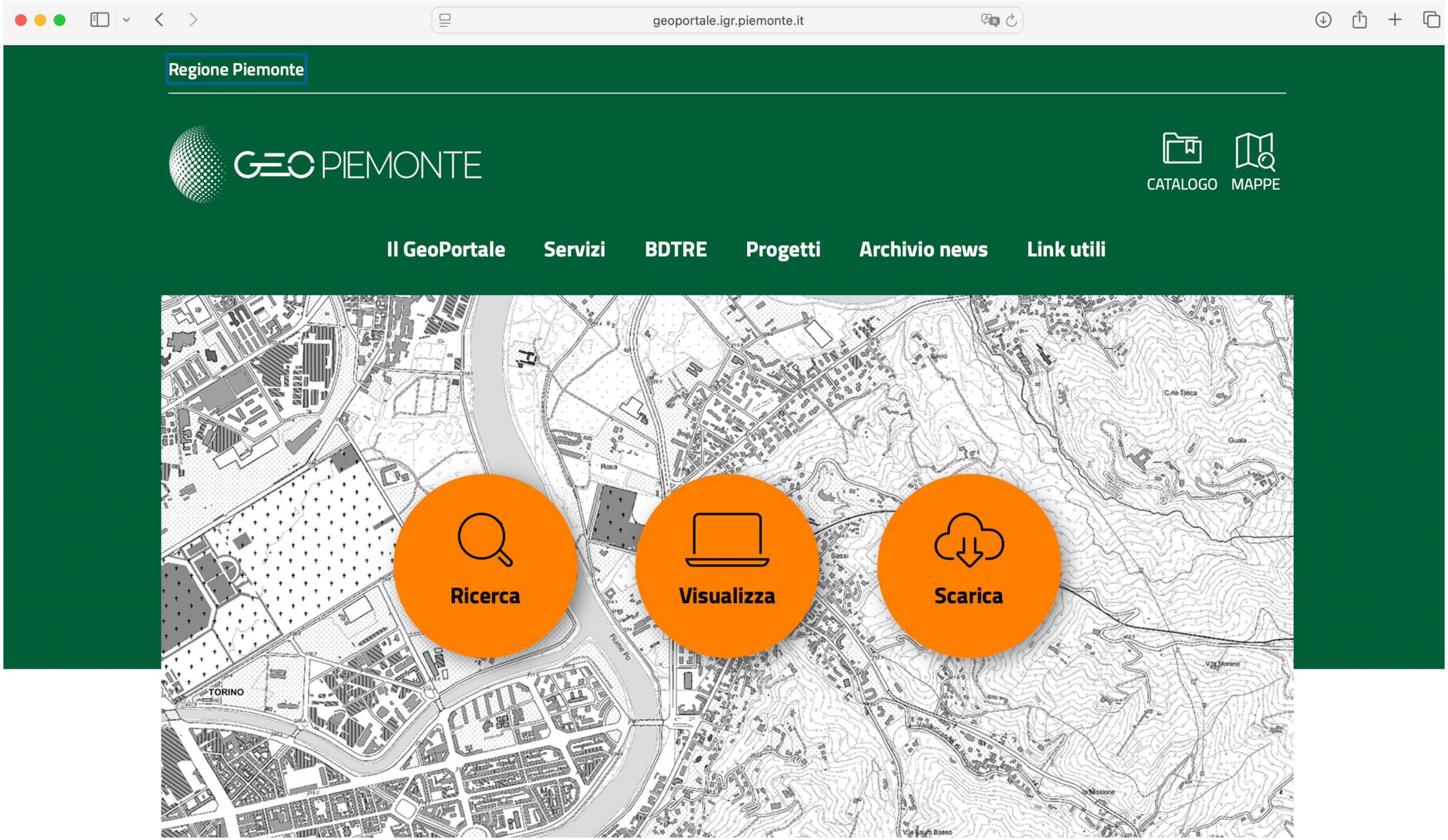
Gives rules, structures, encodings for the management of **interoperable, exchangeable and reusable** 3D city models.

It is a conceptual/logical model.

There are implementations (and data):



- in GML → https://www.ogc.org/standards/citygml/
- In JSON → https://www.cityjson.org


geoportale.igr.piemonte.it
Regione Piemonte
GEO PIEMONTE
CATALOGO
MAPPE
Il GeoPortale
Servizi
BDTRE
Progetti
Archivio news
Link utili
Ricerca
Visualizza
Scarica
TORINO

OpenStreetMap is built by a community of mappers that contribute and maintain data about roads, trails, cafés, railway stations, and much more, all over the world.

Local Knowledge

OpenStreetMap emphasizes local knowledge. Contributors use aerial imagery, GPS devices, and low-tech field maps to verify that OSM is accurate and up to date.

Community Driven

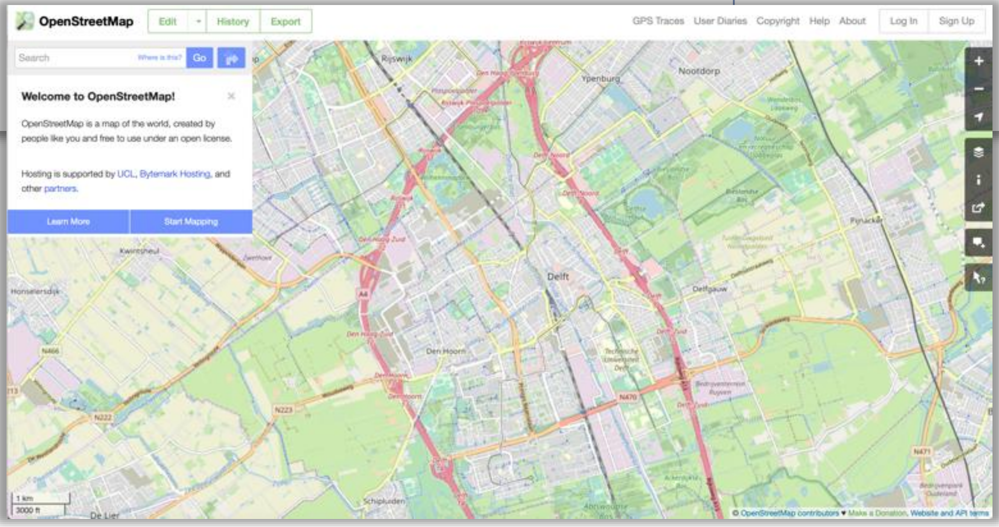
OpenStreetMap's community is diverse, passionate, and growing every day. Our contributors include enthusiast mappers, GIS professionals, engineers running the OSM servers, humanitarians mapping disaster-affected areas, and many more. To learn more about the community, see the OpenStreetMap Blog, user diaries, community blogs, and the OSM Foundation website.

Open Data

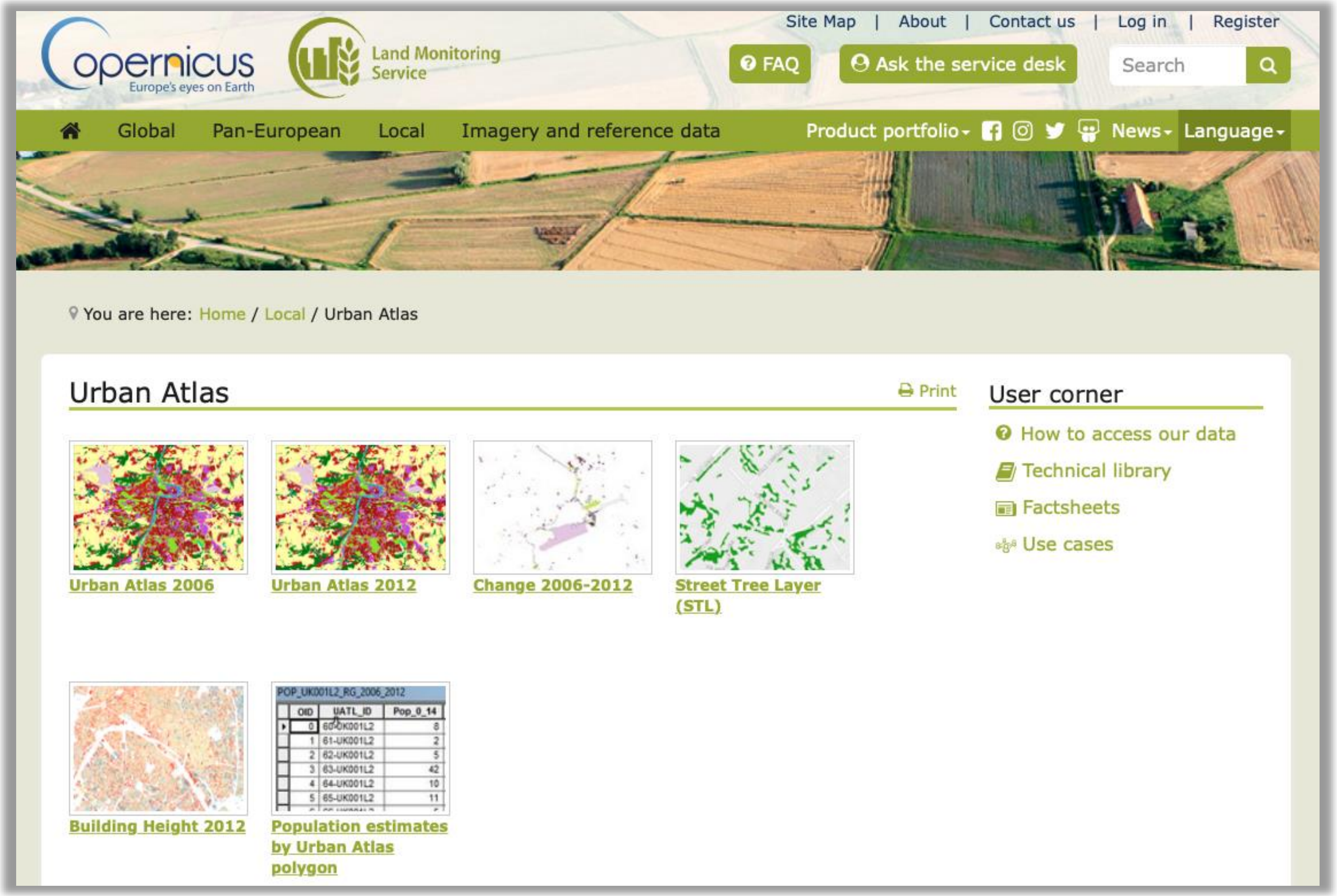
OpenStreetMap is *open data*: you are free to use it for any purpose as long as you credit OpenStreetMap and its contributors. If you alter or build upon the data in certain ways, you may distribute the result only under the same licence. See the Copyright and License page for details.

https://www.openstreetmap.org/#map=8/52.154/5.295

https://www.openstreetmap.nl



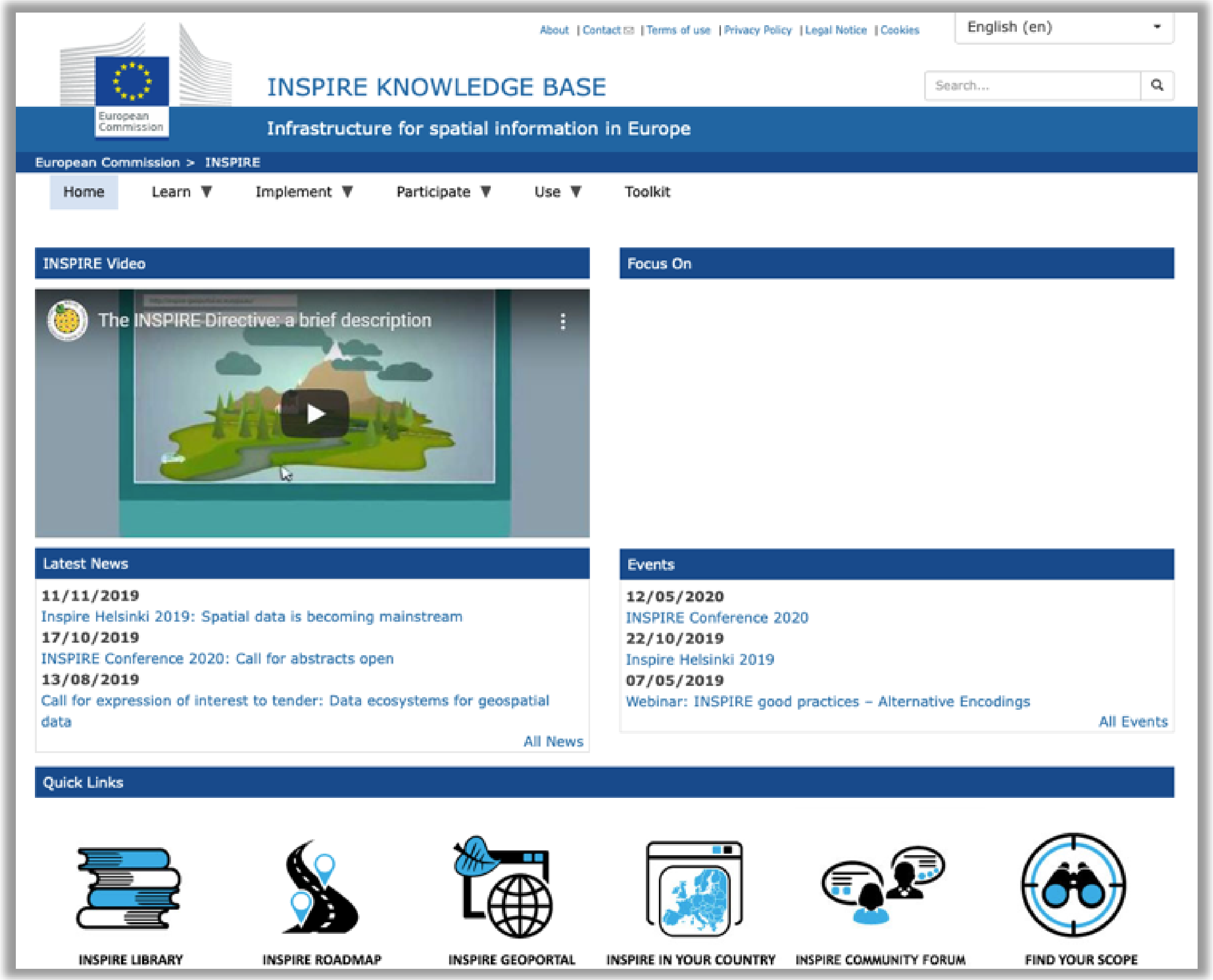
Copernicus is the European Union's Earth Observation Programme, looking at our planet and its environment for the ultimate benefit of all European citizens. It offers information services based on **satellite Earth Observation and in situ (non-space) data**.



https://www.copernicus.eu/en/about-copernicus

https://land.copernicus.eu/local/urban-atlas

The **INSPIRE Directive** aims to create a European Union spatial data infrastructure for the purposes of EU environmental policies and policies or activities which may have an impact on the environment. This European Spatial Data Infrastructure will enable the sharing of environmental spatial information among public sector organisations, facilitate public access to spatial information across Europe and assist in policy-making across boundaries.



https://inspire.ec.europa.eu

Focus1: Dati e sistemi informativi spaziali

Building **I**nformation **M**odel

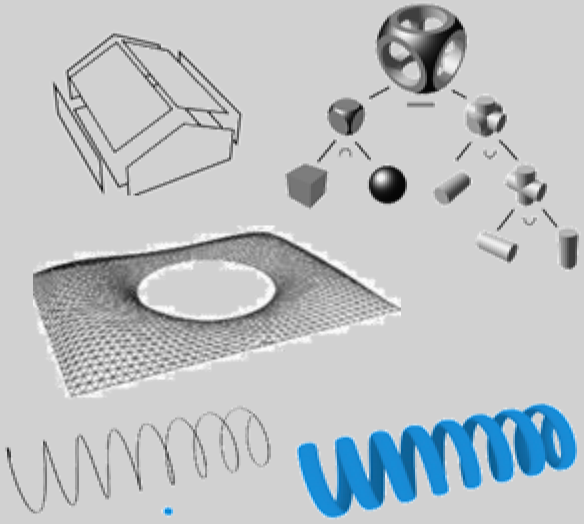
Building Information Models

Evolution of **CAD**;

In **Architecture Engineering and Construction** field (pushed by industry!);

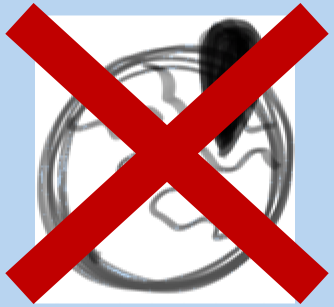
Very high detail objects (e.g. **~1 mm** accuracy);

Geometry

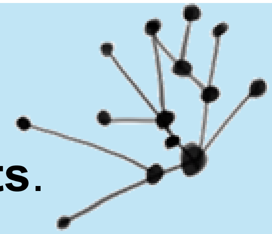


Many possibilities (explicit, Construction Solid Geom., extrusion…)

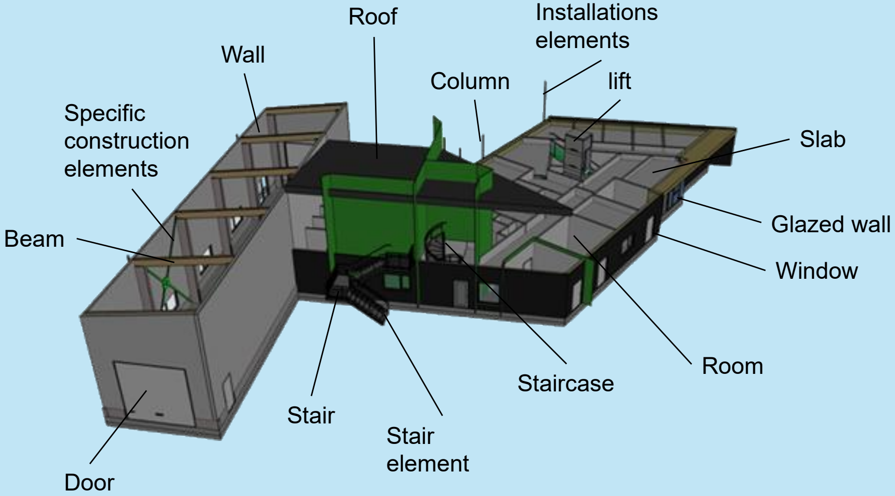
Georeferencing

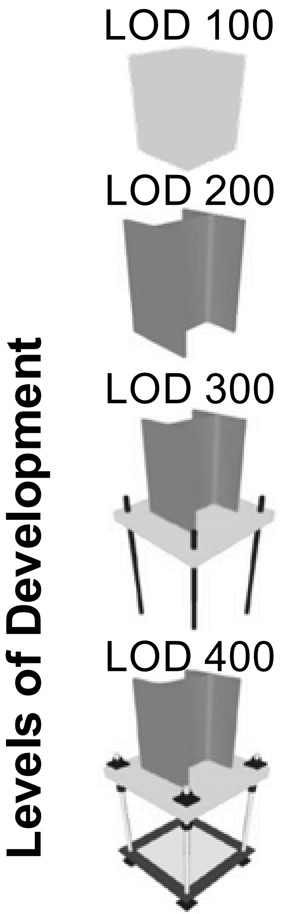


Semantics

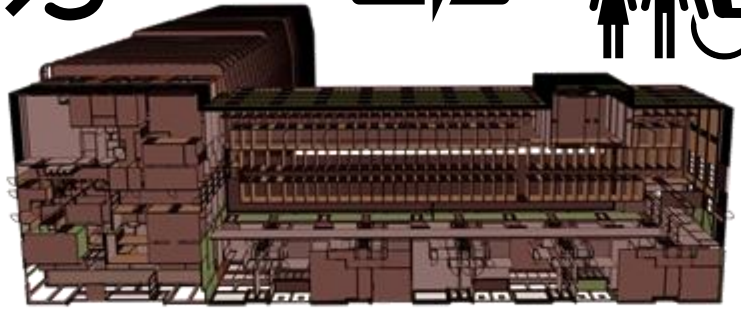


Detailed building **components**.
+
Building **management**: energy information, cost, installations…

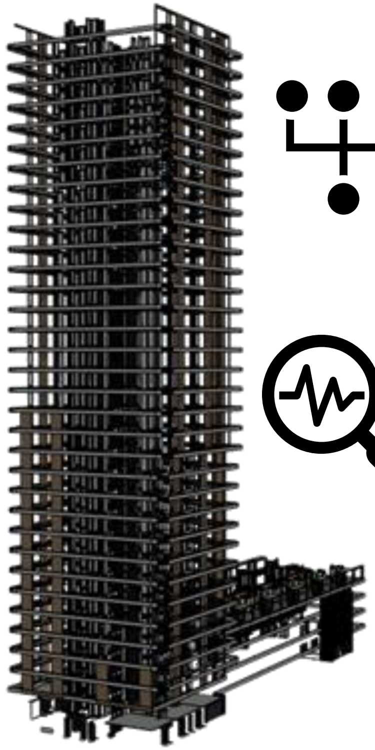




Building Information Models use cases



- **design** options assessment;
- **quantities** and **cost** estimation;
- construction simulation;
- **energy** modelling;
- **project management support** (efficient collaboration, multi-disciplinary project team);
- **facilities and asset management;**
- better **design and construction coordination**;
- **reduced construction costs** (less delays on-site, rework…)
- **reduced operational costs** (seamless information delivery for facilities management at handover).

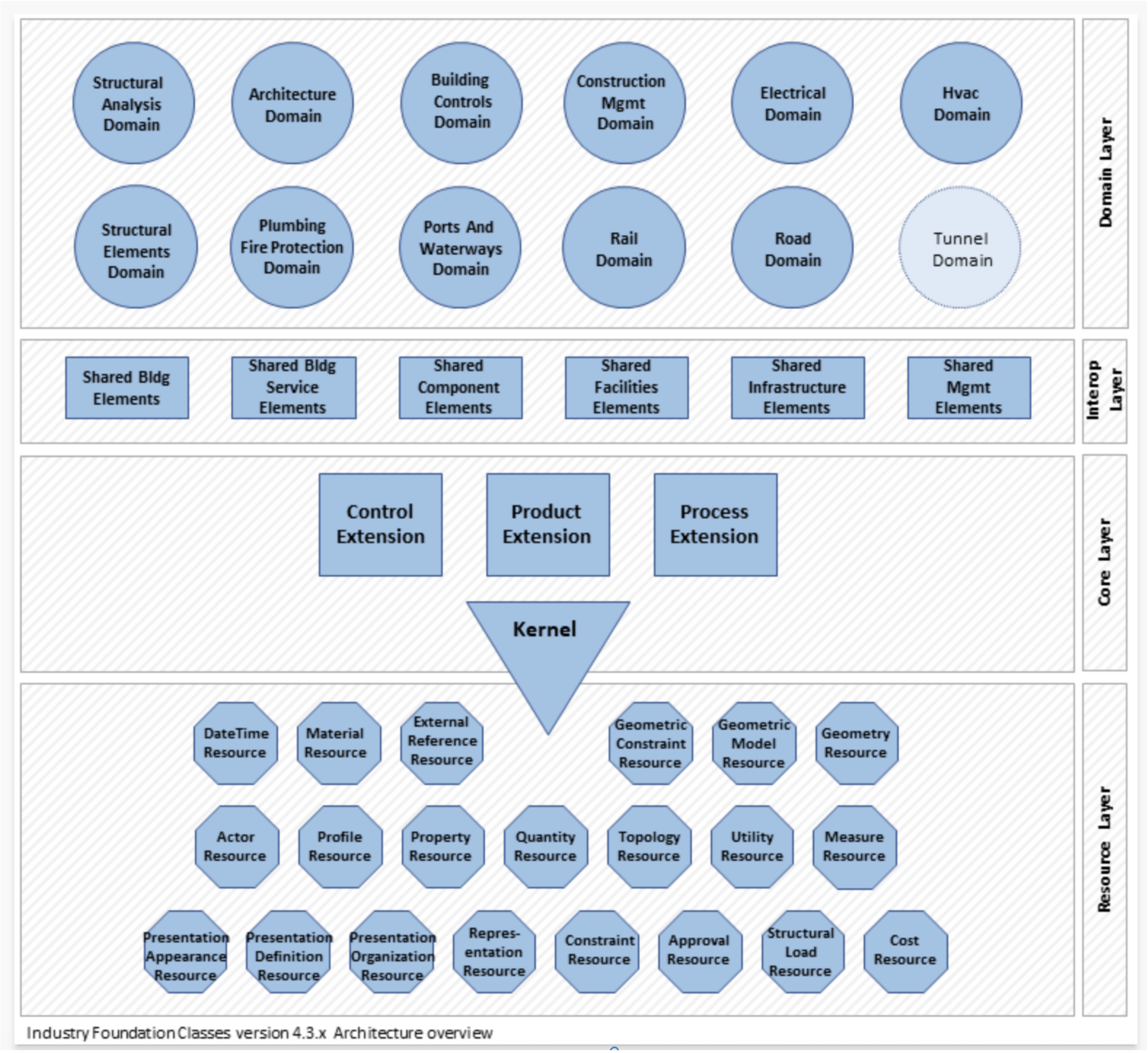


Industry Foundation Classes (IFC) – open standard for BIM

by buildingSMART (ISO 16739).

Part of complex standard about processes, technical requirements and coordination.

https://technical.buildingsmart.org/standards/ifc/

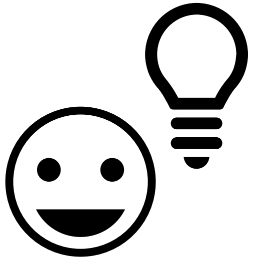


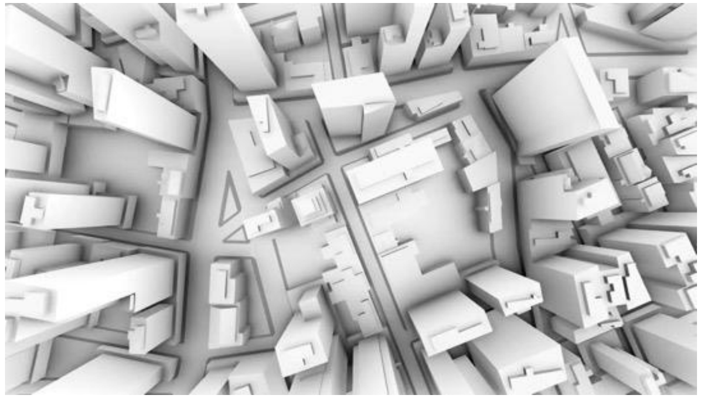
GeoBIM = integration of geoinformation with BIMs

3D geoinformation: **3D city models**

\+

Building Information Models





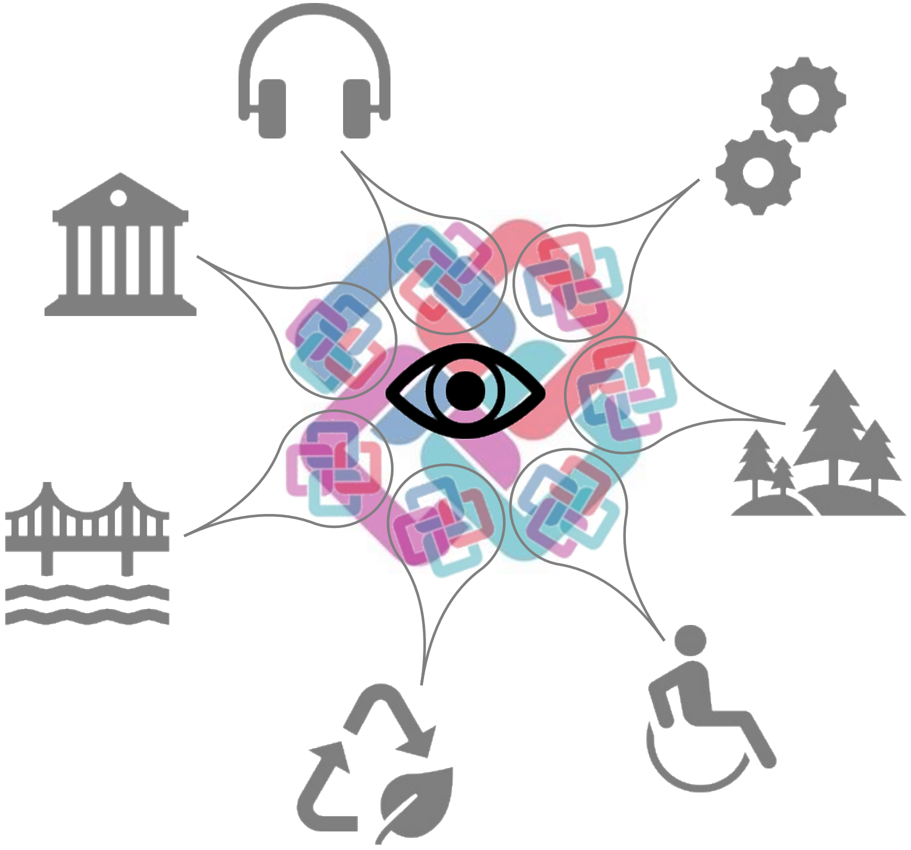
GEO world point of view

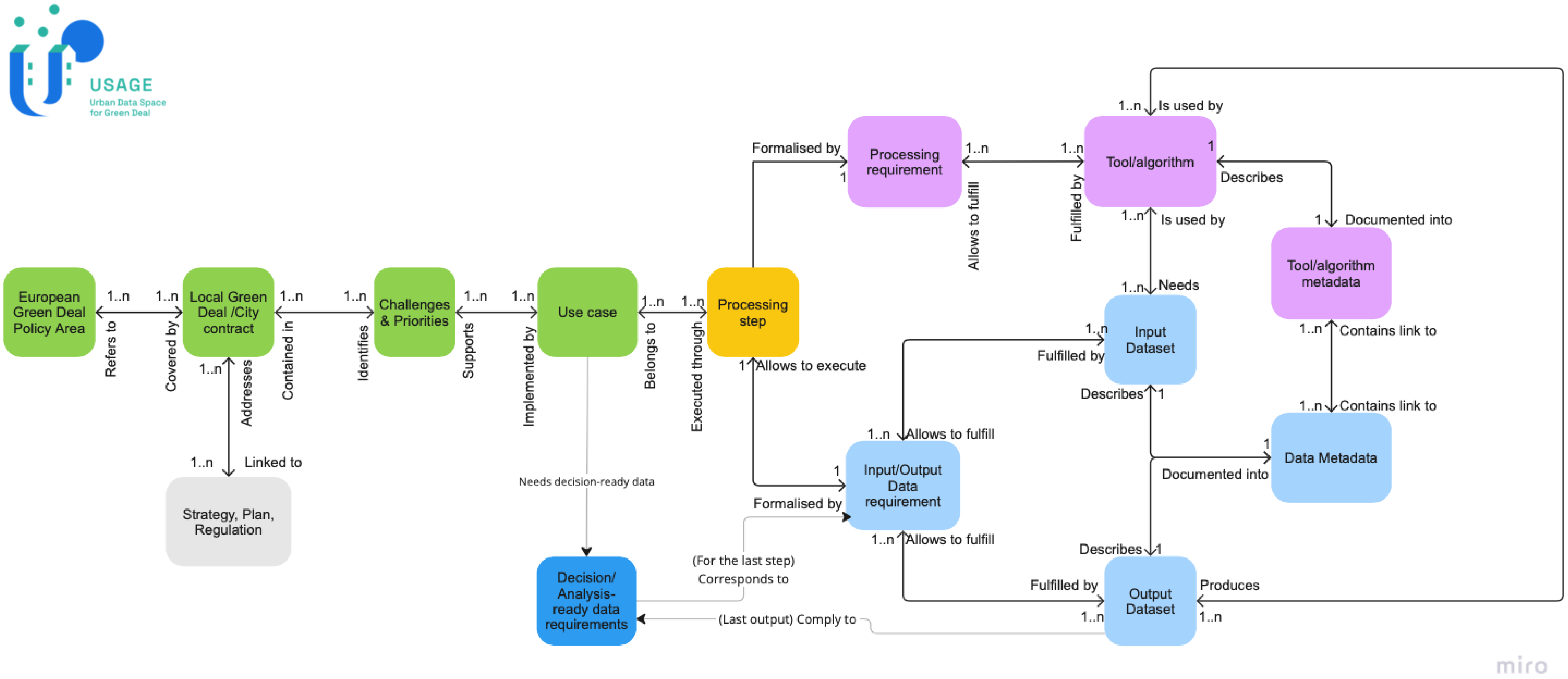
- High **level of detail** 3D cadaster
- **No tasks duplication** (3D data collection)
- Efficient databases **updates** without additional costs
- Effective **data exchange** with **professionals** (architects, engineers, environmental scientists, etc.)
- Stronger information for **lifecycle asset management & city analysis**

BIM world point of view

- **Context** for design reference
- Improved **test of building properties**: designed building into its context
- Test of the **impact of the building** on the city or landscape.
- **Multiscale** vision (from construction elements to whole territories)

Focus2: Definizione di casi d'uso e requisiti




USAGE
Urban Data Space for Green Deal
European Green Deal Policy Area
Refers to
1..n
Covered by
1..n
Local Green Deal /City contract
Contained in
1..n
Addresses
1..n
Linked to
1..n
Strategy, Plan, Regulation
Identifies
1..n
Challenges & Priorities
Supports
1..n
Implemented by
1..n
Use case
Belongs to
1..n
Executed through
1..n
Processing step
Formalised by
1
Processing requirement
Allows to fulfill
1..n
Fulfilled by
1..n
Tool/algorithm
Is used by
1..n
1
Describes
Documented into
1
Tool/algorithm metadata
Contains link to
1..n
Needs
1..n
Input Dataset
Fulfilled by
1..n
Describes
1
Data Metadata
Documented into
1
Allows to execute
1
Input/Output Data requirement
Allows to fulfill
1..n
Formalised by
1
Needs decision-ready data
Decision/ Analysis-ready data requirements
(For the last step) Corresponds to
(Last output) Comply to
Output Dataset
Fulfilled by
1..n
Produces
1..n
Describes
miro

Objective parameters

Integrability Pre-assessment parameters

Scope

Spatial / Geographical extent

Time framework

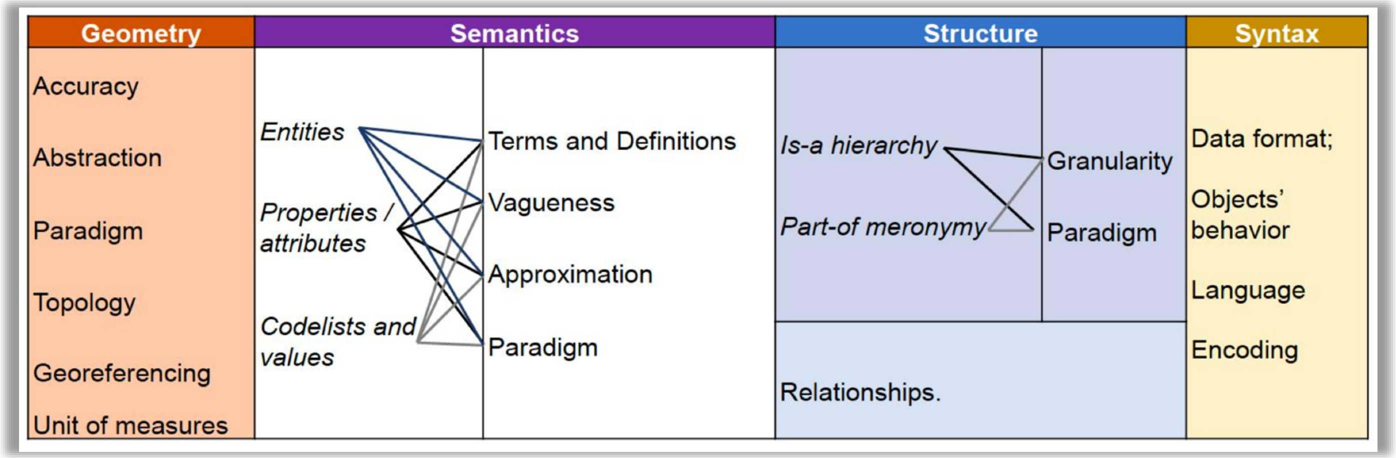
Integrability Detailed assessment parameters

Geometry

Semantics

Structure

Syntax



Qualitative parameters

Additional information supporting data interpretation and re-use

Original goal (intended use)

(Software-specific) implementation details

Authorship and context within which data were generated: discipline, ethno-/cultural-/socio-based view, human cognitive diversity

Lineage

Noardo, F. Multisource Spatial Data Integration for Use Cases Applications. *Transactions in GIS* 2022, 26,7, 2874-2913 (doi: https://doi.org/10.1111/tgis.12987).
EU MSC cofund 'Leading Fellows' Postdocs Programme, grant agreement No. 707404. *Multisource Spatial data integration for smart city applications.*

OGC
Data Exchange Toolkit

Standard data model profiling tool

↑↓

Data requirements specification template

- **Explain data requirements**
- Validate data against standard-based data requirements

Dataset requirements

Ref	GeoDCAT										-	ISO19131
Field	Title	Spatial / geographic coverage	Temporal coverage	Description	Access rights / rights holder (Licence)				Geometry: Reference system	Geometry: Unit of measure	Reference (standard) data model	Delivery Format
Prescribed content	*Title of the dataset, or topic, intended to identify the scope*	*Spatial extent of the dataset, defined in the chosen CRS*	*Temporal framework of the data representation*	*Short description of the dataset*	*BY*	*SA*	*NC*	*ND*	*Coordinate Reference System EPSG - OGC ID in Rainbow*	*Unit of measure*	*Reference data model and version URI in Rainbow (to be selected via a codelist name – URI)*	*Encoding format of the dataset*
Example	Example Building Permit 3D city model	surrounding of the new building e.g. parcel in which the new building is designed + 1km buffer	Updated with latest and approved new works	3D city model LoD 3.3 intended to support automatic checks of urban regulation for new buildings within a GIS environment.	Yes	Yes	No	No	Local projected CRS	m	CityJSON v.1.1 URI	JSON

Objects requirements

Ref	-	Profile of standard data model					-	GeoDCAT	-
Field	Object	Reference data model subset	Class	SuperClass with property	Property	Geometry object	Accuracy	Spatial Resolution / LoD	Additional notes on the spatial representation
Prescribed content	*Mapped city object*	*URI in Rainbow (to be selected via a codelist name – URI)*	*Selected from drop-down list based on the standard data model (with URIs in Rainbow)*	*Selected from drop-down list based on the standard data model (with URIs in Rainbow)*	*Selected from drop-down list based on the standard data model (with URIs in Rainbow)*	*Selected from drop-down list based on the standard data model (with URIs in Rainbow)*	*Based from the use case requirements defined with stakeholders*	*According to Biljecki et al., 2016*	*Further specifying the kind of representation needed, based on the use case requirements*
Example	Existing buildings	Building	Building	Building	Geometry.type	MultiSurface	0.2 m	LoD 3.3	To be included: external walls with windows, all protruding parts of the building, height = max height of the building at the top of the roof
		Building	Window	Building Building.children*	Geometry.type	MultiSurface	0.2 m		
		Building	WindowSurface	AbstractThematicSurface	boundary	GM_MultiSurface	0.2 m		
		Building	WallSurface	AbstractThematicSurface	boundary	GM_MultiSurface	0.2 m		
		Building	WallSurface	AbstractThematicSurface	lod3MultiSurface	GM_MultiSurface	0.2 m		
	Road, including sidewaks, flowerbed parkings	Road	Road	Road	Geometry.type	MultiSurface	0.2 m	LoD0	
		Road	TrafficArea	AbstractThematicSurface	lod0MultiSurface	GM_MultiSurface	0.2 m	LoD0	
		Road	AuxiliaryTrafficArea	AbstractThematicSurface	lod0MultiSurface	GM_MultiSurface	0.2 m	LoD0	

CityJSON

CityGML

INSPIRE Data model

Francesca Noardo (OGC), Piergiorgio Cipriano (DedaNext)

		ISO19115	INSPIRE[21]	IFC	CityJSON
Contents and procedures	*Spatial extent*	Geographic location of the dataset	geographic bounding box		geographicalExtent
	Temporal frame	Dataset reference date, Additional extent information for the dataset	temporal reference, temporal extent, date of last revision, date of creation		datasetReference Date
	Scope	Dataset topic category	resource type, topic category	IDM, IDS	datasetTopicCategory
	Goal and use case requirements	Abstract describing the dataset	Resource abstract	IDM, IDS	Abstract, specificUsage
	Lineage	Lineage	Lineage		lineage
	Author	Dataset responsible party	Responsible organization		
	Implementation requirements		specifications[22]		spatialRepresentationType
Geometry	*Accuracy*		specifications		
	Abstraction level	Spatial resolution of the dataset	Spatial resolution		presentLoDs
	Geometry paradigm	Spatial representation type	specifications	IDM, IDS, MVD	
	Topology		specifications	IDM, IDS, MVD	
	Georeferencing	Reference system	specifications	IfcMapConversion	referenceSystem
	Unit of measure		specifications	IfcProject - ProjectUnits	
Semantics	*Entities*		specifications	IDM, IDS, MVD	
	Properties and attributes		specifications	IDM, IDS, MVD	
	Codelists and values		specifications	IDM, IDS, MVD	
	Terms		specifications	IDM, IDS, MVD	
	Accuracy (vagueness)		specifications		
	Approximation level		specifications		
	Semantic paradigm		specifications	IDM, IDS, MVD	
	Language	Dataset language	resource language		datasetLanguage
	Encoding		specifications		
Structure	*Is-a hierarchies;*		specifications	IDM, IDS, MVD	
	Part-of meronymic hierarchies;		specifications	IDM, IDS, MVD	
	Relationships		specifications	IDM, IDS, MVD	
	Reference data model,		specifications	IDM, IDS, MVD	
	version		specifications	IDM, IDS, MVD	
	profile		specifications	IDM, IDS, MVD	
	extensions		specifications	IDM, IDS, MVD	
	Granularity		specifications	IDM, IDS, MVD	
S	*Data format*		specifications	IDM, IDS, MVD	
	Objects' behavior		specifications	IDM, IDS, MVD	

Initial simple rules…

Good **metadata**

Specify data **requirements**

Use Open standards as much as possible

Align with:

- User **requirements**
- Use case **requirement**
- Implementation **requirements**

Focus3: Risultati di progetti per:

- Permessi di costruire digitali

CHEK
DIGITAL BUILDING PERMIT

Change toolkit for
Digital Building Permit
https://chekdbp.eu

- Green Deal - Sostenibilità ambientale / transizione ecologica

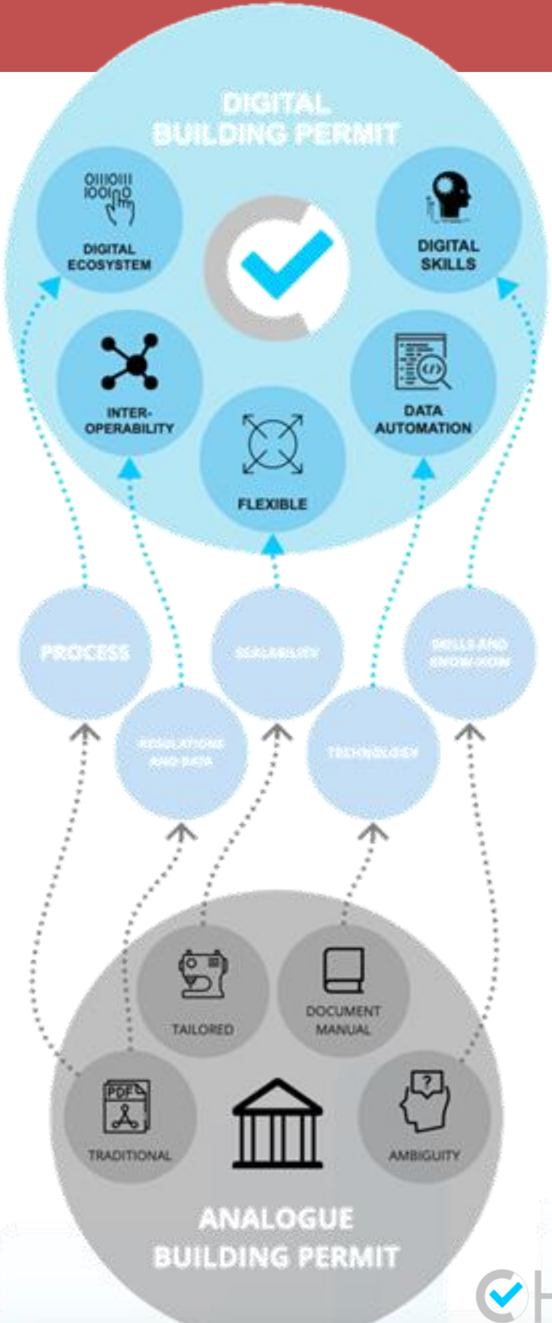
USAGE
Urban Data Space
for Green Deal

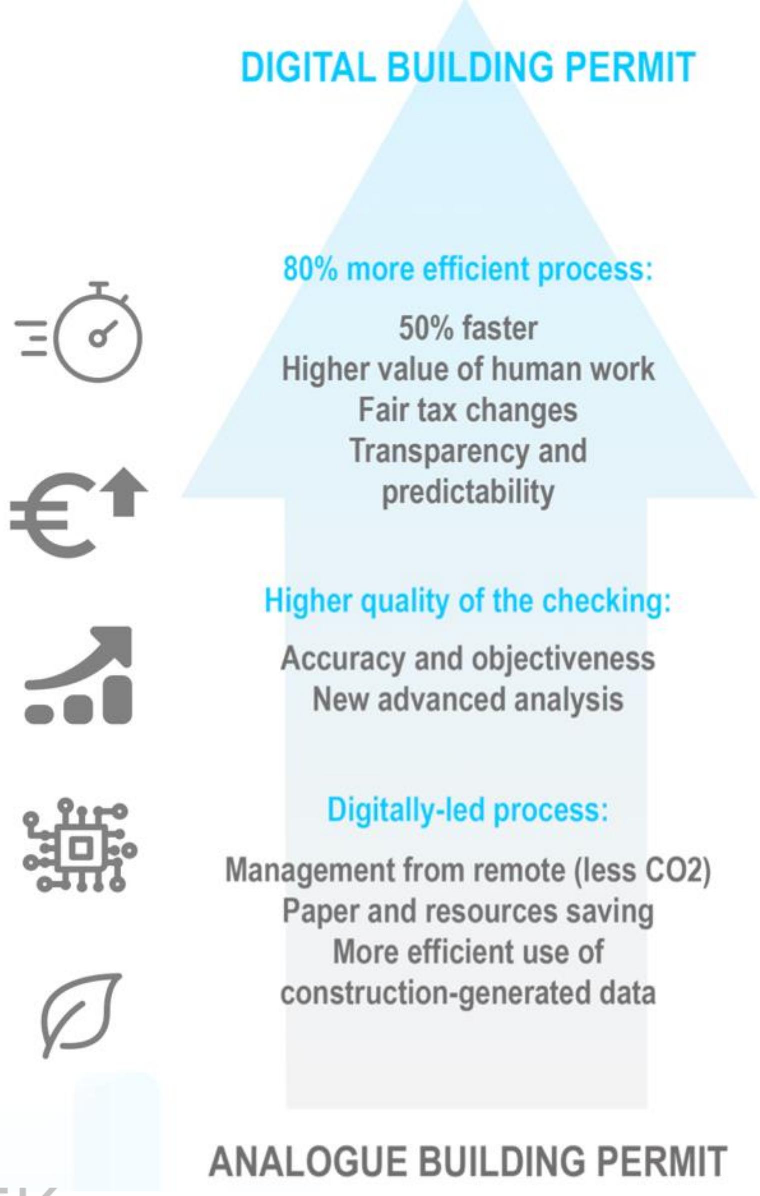
https://www.usage-project.eu/home

Permessi di costruire digitali

Cos'è un permesso di costruire digitale?

La digitalizzazione del rilascio dei titoli abitativi implica il passaggio da sistemi (analogici) basati su documenti a processi digitali basati su dati (Building Information Models, 3D city models, Sistemi Informativi Territoriali).





CHEK DIGITAL BUILDING PERMIT

https://chekdbp.eu/

Adozione del BIM per opere in Italia

OBBLIGO DI ADOZIONE DEL BIM
nelle opere pubbliche

BibLus-net

TIPOLOGIA DI LAVORI	IMPORTO	DATA
Lavori complessi	≥ **100 milioni** di euro	dal 1° gennaio 2019
	≥ **50 milioni** di euro	dal 1° gennaio 2020
	≥ **15 milioni** di euro	dal 1° gennaio 2021
Opere di **nuova costruzione** ed interventi su **costruzioni esistenti**, fatta **eccezione per le opere di manutenzione ordinaria**	≥ **15 milioni** di euro ~~(≥ 5,35 milioni di euro)~~	dal 1° gennaio 2022
Opere di **nuova costruzione** ed interventi su **costruzioni esistenti**, fatta **eccezione per le opere di manutenzione ordinaria e straordinaria**	≥ **5,35 milioni** di euro ~~(≥ 1 milione di euro)~~	dal 1° gennaio 2023
Opere di **nuova costruzione** ed interventi su **costruzioni esistenti**, fatta **eccezione per le opere di manutenzione ordinaria e straordinaria**	≥ **1 milione** di euro ~~(≤ 1 milione di euro)~~	dal 1° gennaio 2025

Directive 2014/24/EU of the European Parliament and of the Council on public procurement.

decreto 2 agosto 2021, n. 312

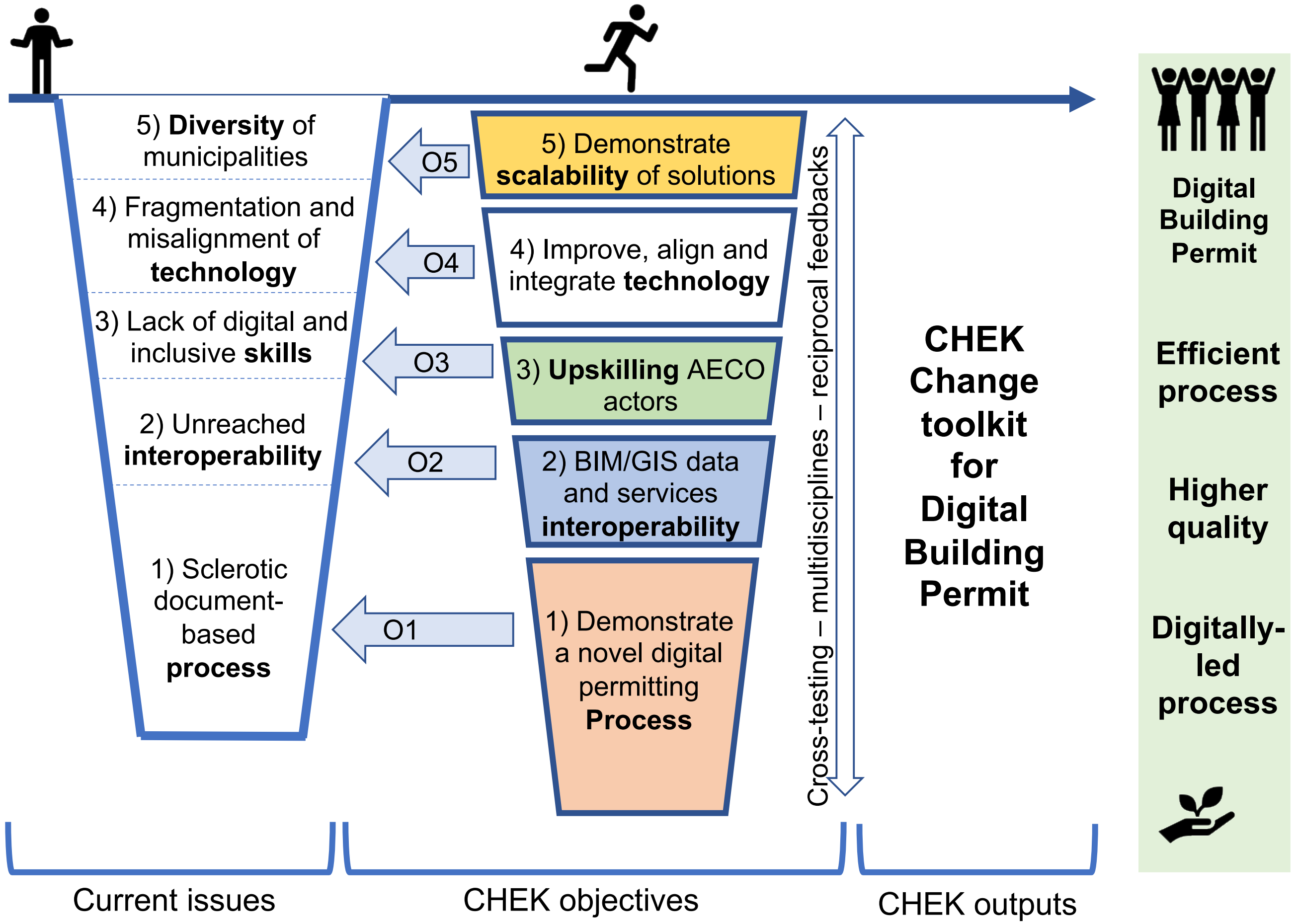
modifica il precedente

dm 560/2017

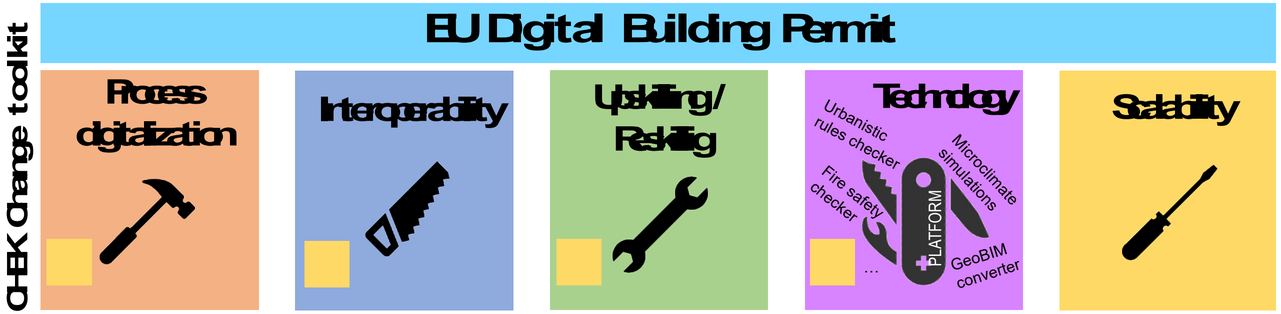
Francesca Noardo (OGC), Piergiorgio Cipriano (DedaNext)

Open Geospatial Consortium
deda.next envision public services

Introduction CHEK



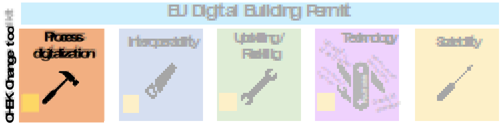
Process change – Information requirements – GeoBIM – OpenAPI-based microservices – Training



CHEK
DIGITAL BUILDING PERMIT

Co-funded by the European Union

This project has received funding from the European Union's Horizon Europe programme under Grant Agreement No.101058559



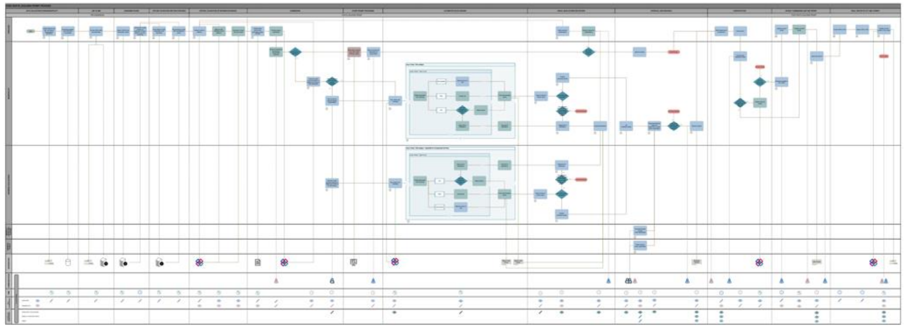
Objective 1: Un nuovo processo di autorizzazione edilizia digitale

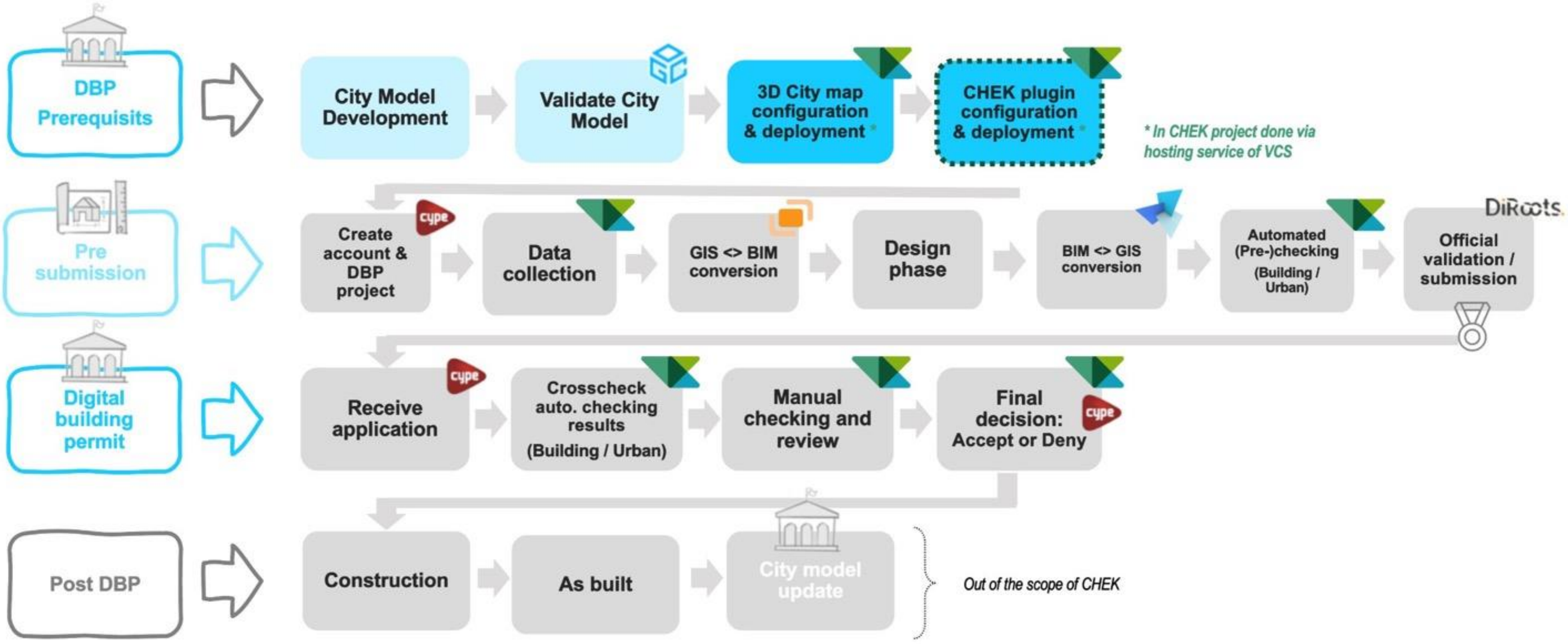
Definizione di un **nuovo processo**, da:

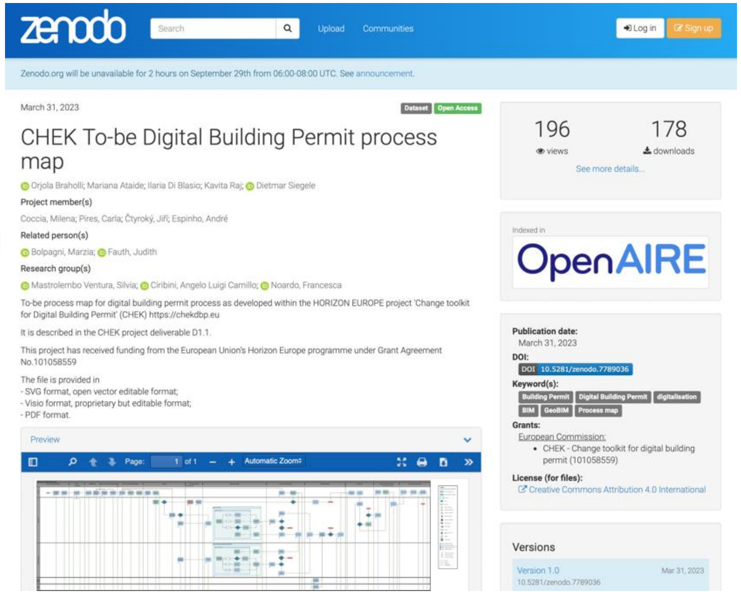
Pratiche attuali,

Coinvolgimento diretto funzionari pubblici,

Uso di 3D city models e BIM





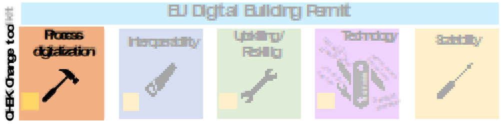


CHEK DIGITAL BUILDING PERMIT

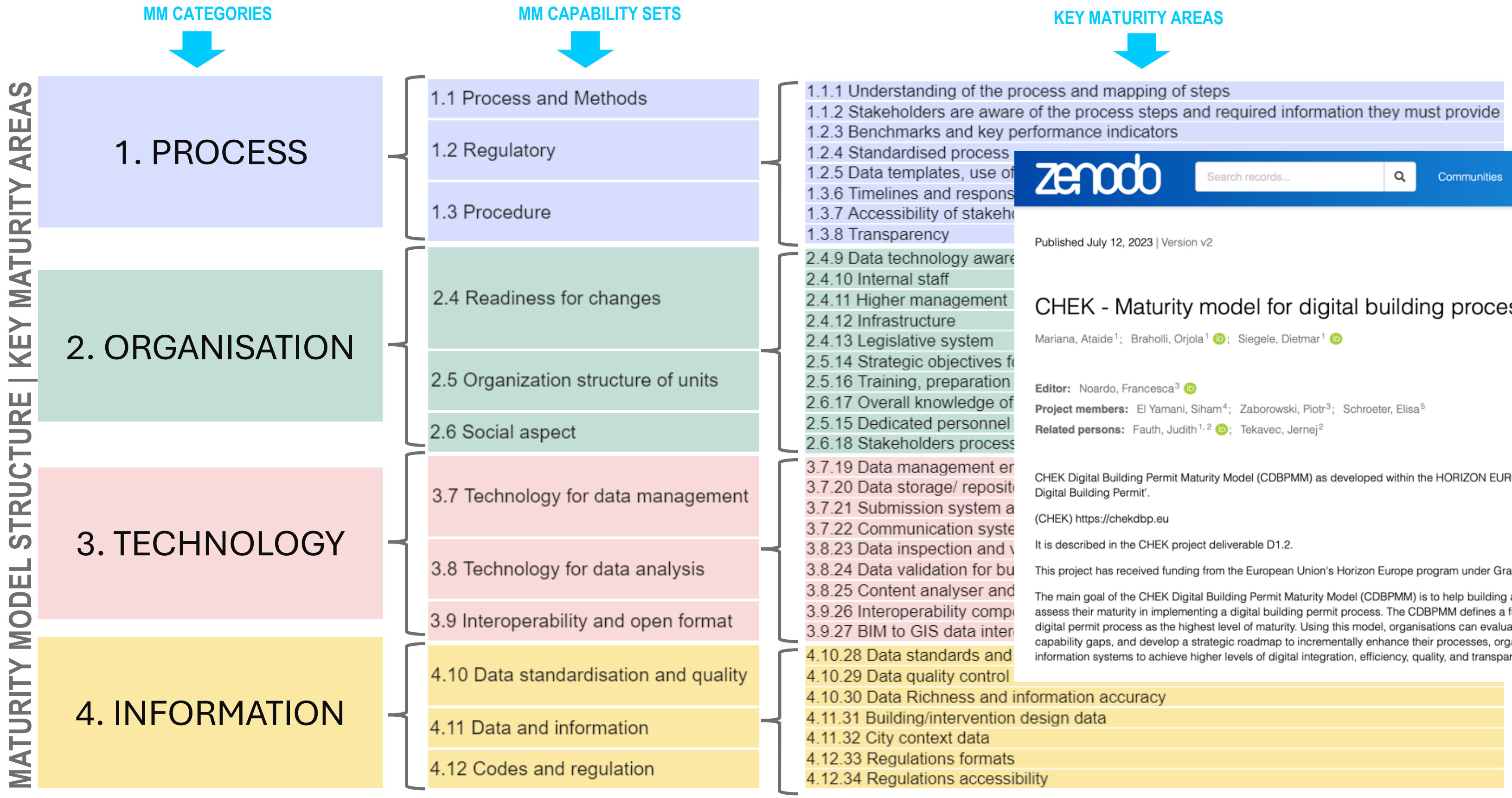
https://chekdbp.eu

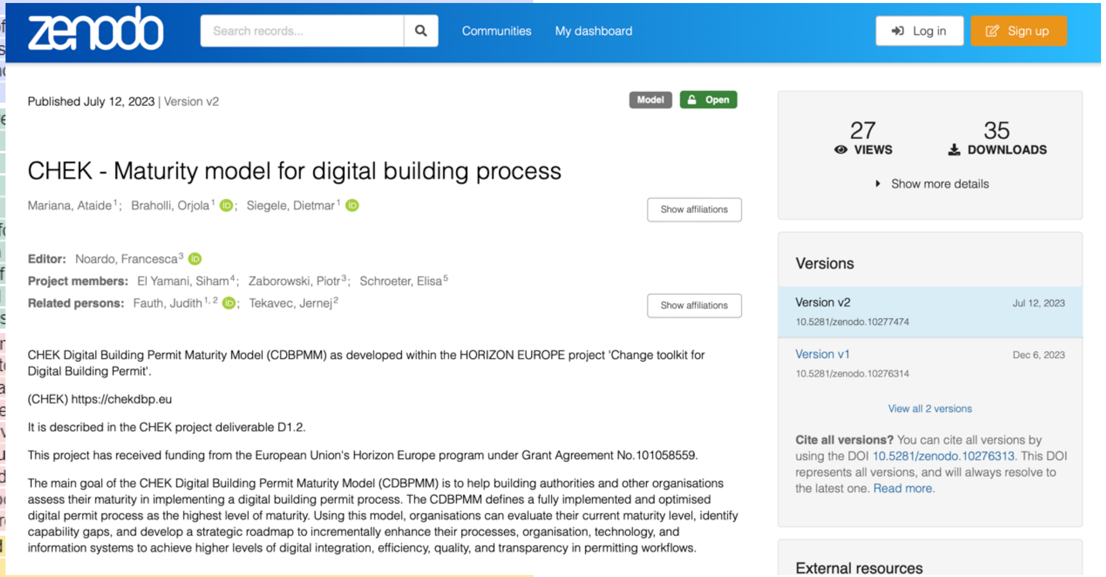
Funded by the European Union

Objective 1: Un nuovo processo di autorizzazione edilizia digitale



Definizione di un **Maturity Model e roadmap**, riferimento nella valutazione del livello corrente di digitalizzazione e support alla pianificazione futura

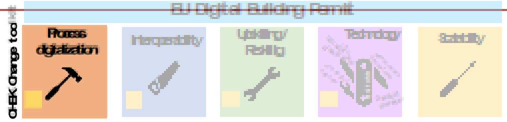




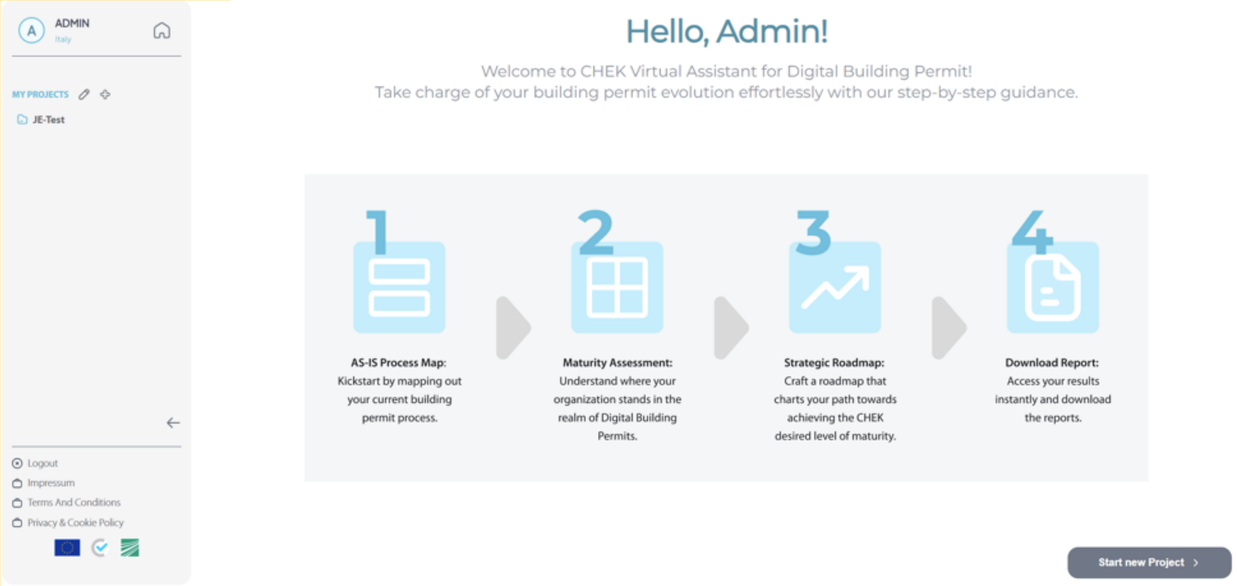
CHEK DIGITAL BUILDING PERMIT

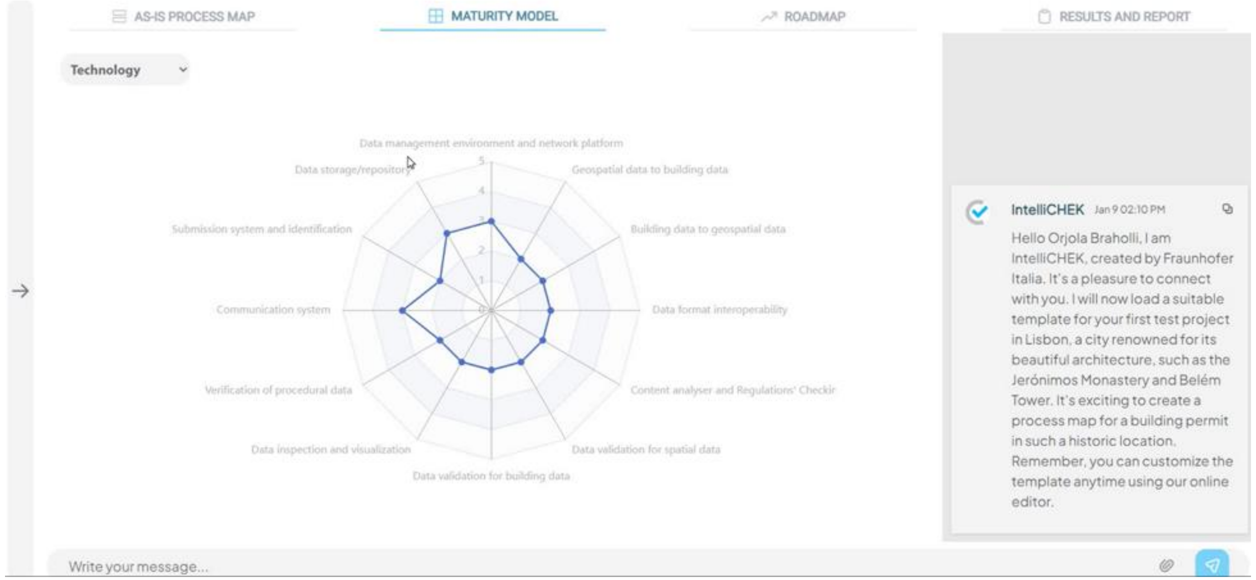
https://chekdbp.eu

Objective 1: Un nuovo processo di autorizzazione edilizia digitale



Sviluppo del **CHEK Change management Virtual Assistant** (con Intelligenza Artificiale) a supporto dell'utilizzo degli strumenti forniti dal Progetto per definire e valutare il livello corrente di digitalizzazione e supportare la pianificazione futura





Objective 2: Interoperabilità

Interpretazione **regolamenti edilizi** (possibili strumenti AI; parzialmente efficaci)

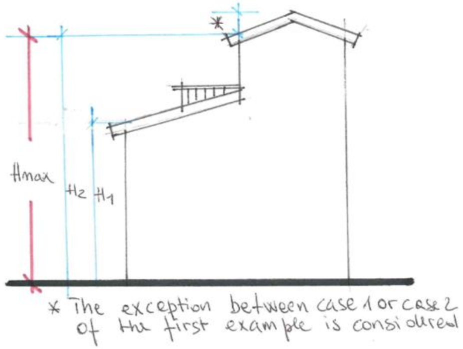
Definizione **Regulatory Information Requirements**

Definizione dei requisiti dei dati (GIS e BIM) basati su **standards**

Normative clause (ITA language)
(Regolamento edilizio comunale of the Municipality of Ascoli Piceno – art. 61)

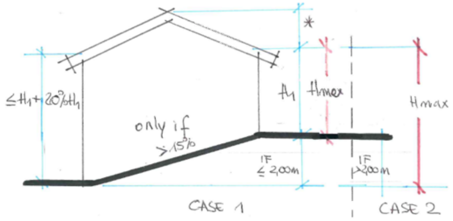
Comma 4 - Le distanze minime tra i fabbricati fra i quali siano interposte strade, con esclusione della viabilità a fondo cieco al servizio di singoli edifici o insediamenti, debbono corrispondere alla larghezza della strada maggiorata di:
(1) ml.5.00 per lato, per strade di larghezza inferiore a ml. 7.00;
(2) ml.7.50 per lato, per strade di larghezza compresa tra ml. 7.00 e ml. 15.00;
(3) ml. 10.00 per lato, per strade di larghezza superiore a ml. 15.00.

Height of facades: This is the height of each part of the elevation into which the building can be broken down, including setbacks, measured from the *ground line* to the *roof line*.



The ground line is defined by the intersection of the wall of the elevation with the street level or sidewalk level or ground level at final settlement.

(3) For buildings on land with a natural slope of more than 15%, the maximum height permitted by the town planning instruments, unless more restrictive prescriptions of the same, may be exceeded by 20% in the downstream parts of the facade, with an absolute maximum of 2.00 (see figure 7). [i.e. if slope > 15%, Maximum height can be 20% higher in the lowest part of the façade, but never more than 2 m.]



RMUEL Art. 41, Paragraph 3

Text

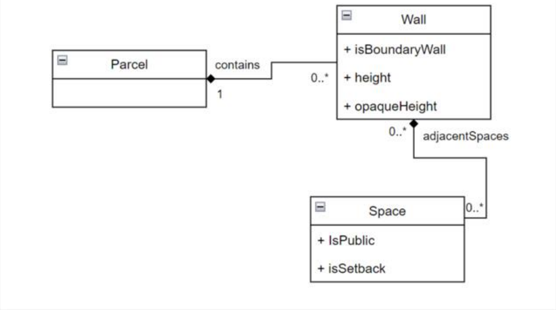
3 - Facing the public spaces, boundary walls may not exceed 1.20 metres in height, extending to the boundary wall on the side of the parcel in the part corresponding to the building's setback, where this exists, and boundary walls up to 2.0 metres in height are permitted, when supplemented with hedgerow.

Pseudocode

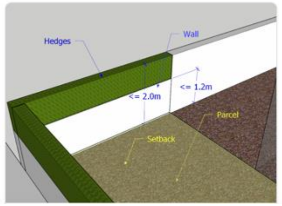
FOREACH BoundaryWall IN Parcel:
IF isFacing(BoundaryWall, PublicSpace) OR isFacing(BoundaryWall, Setback):
CHECK IF BoundaryWall *height* <= 1.2 *m*
IF FAIL:
CHEK IF *opaqueHeight*(BoundaryWall) <= 1.2 *m*

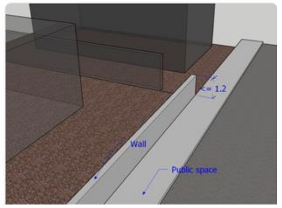
CHEK IF BoundaryWall *height* <= 2.0 *m*

ASK APPROVED "boundary walls up to 2.0 metres in height are permitted, when supplemented with hedges."



Image





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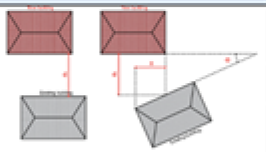
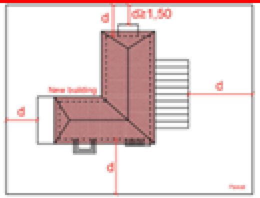
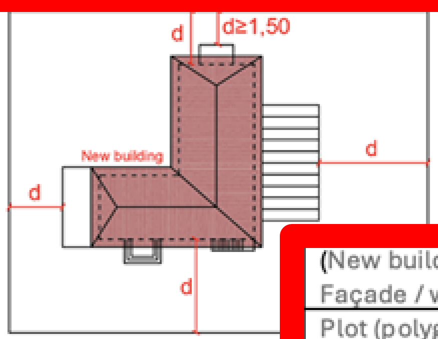
Regulation: Distances

Definitions - Regolamento edilizio comunale of the Municipality of Ascoli Piceno – Art. 13	Clarifications by municipality technicians (informal)	In New Building	In Context
Paragraph o – Building-building distance: Is the minimum distance between walls or building volumes facing the new building (considering all the building volumes and protrusions) except for the walls facing the building *interior spaces* Two walls are considered to be facing each other, when the angle formed by the extension of the respective façade projections on the plane is lower than 70 degrees sexagesimal and their overlap (i.e. orthogonal projection on each other) is higher than 1/4 of the minimum distance between the walls themselves. For graded buildings (i.e. buildings which are composed by horizontally or vertically articulated volumes), the distance is measured at each setback.	 The sentence "For graded buildings (i.e. buildings which are composed by horizontally or vertically articulated volumes), the distance is measured at each setback." Just reinforces the idea that the minimum distance is considered among any possible protruding volume of the building, in any position of the façade.	Walls	(Existing in the context) Walls/Buildings
		Façade	(New building) Façade / wall surface
		Building Interior space? (Patio, yard, cloister)	Building Interior space? (Patio, yard, cloister)
Paragraph p – **Building- boundaries distance.** It is the minimum distance between the vertical projection of the wall of the building and the boundary of the *plot* (considering all the building volumes and protrusions), **Border:** - separation line of the different existing ownerships, - line defining the different *plots or compartments* of the implementation plans, - bordering line of public areas for services or equipment, identified in urban planning instruments.	 Public areas for services: see *primary and secondary infrastructure* above	Façade	(New building) Façade / wall surface
			Plot (polygon)
			Cadastral parcels (ownership)
			Implementation plans plots / compartments
		Road	Road
		Public parking	Public parking
		Utilities (sewerage, water supply, electricity and gas network)	Utilities (sewerage, water supply, electricity and gas network)

Paragraph p – **Building- boundaries distance.** It is the minimum distance between the vertical projection of the wall of the building and the boundary of the *plot* (considering all the building volumes and protrusions),

Border:
- separation line of the different existing ownerships,
- line defining the different *plots or compartments* of the implementation plans,
- bordering line of public areas for services or equipment, identified in urban planning instruments.



Public areas for services
and secondary infrastru

(New building)
Façade / wall surface
Plot (polygon)
Cadastral parcels (ownership)
Implementation plans plots / compartments
Road
Public parking
Utilities (sewerage, water supply, electricity and gas network)

CityGML CityJSON

CHEK CityJSON profile for the municipality of Ascoli Piceno

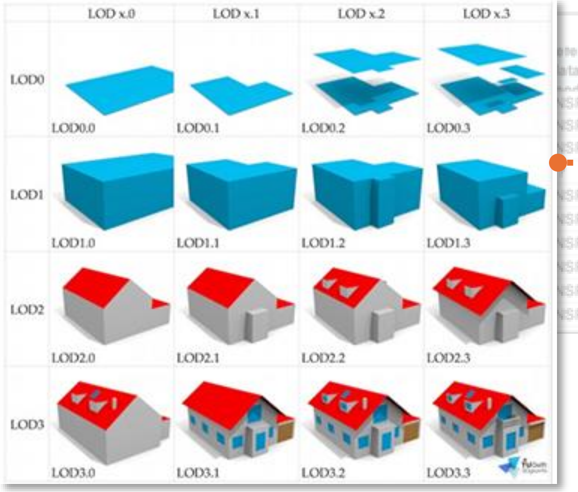
Title	Semantics and Structure --> CityJSON Profile							Geometry				Data format
	CityJSON Class	CityJSON module	CityJSON property	notes	Attribute value type	spatial representaion type	reference data model	Accuracy	Spatial resolution	Reference system	Unit of measure	
Existing building - Function	Building	Building	?		codelist 'APC_BuildingFuncti	-	CityGML v.3	-	-	-	-	JSON
Existing Building - legalHeight	Building	Building Extension?	legalHeight	attribute to host, in the future, the height meas		-						JSON
Existing Building - legalVolume	Building	Building Extension?	legalVolume	attribute to host, in the future, the volume of th		-						JSON
Existing building - Façade	WallSurface	Building	semanticSurface?		-	MultiSurface / CompositeSurface?	CityGML v.3	10 cm	LoD1.3???	projected national CRS	m	JSON
Existing building - Façade	WallSurface	Building	parent	to building	-	-	CityGML v.3	-	-	-	-	JSON
Existing building - Façade - hasWindows	WallSurface	Building Extension?	hasWindows		Boolean	-	CityGML v.3	-	-	-	-	JSON
Road	Road	Transportation	semanticSurface?		-	MultiSurface / CompositeSurface?	CityGML v.3	10 cm	LoD0 (3D)	projected national CRS	m	JSON
Sidewalk	TrafficArea (or Traffic square?)	Transportation	semanticSurface?		-	MultiSurface / CompositeSurface?	CityGML v.3	4 cm	LoD0	projected national CRS	m	JSON
Road furniture areas	AuxiliaryTrafficArea (or TrafficSquare?)	Transportation	semanticSurface?		-	MultiSurface / CompositeSurface?	CityGML v.3	4 cm	LoD0	projected national CRS	m	JSON
Public Parking	AuxiliaryTrafficArea (or TrafficSquare?)	Transportation	semanticSurface?		-	MultiSurface / CompositeSurface?		4 cm	LoD0	projected national CRS	m	JSON
Utilities (sewerage, water supply, electricit	?	?	?	?	?	?		10 cm		projected national CRS		JSON
Green public spaces	PlantCover	Vegetation	lod0MultiSurface??		-	MultiSurface / CompositeSurface?	CityGML v.3	10 cm	LoD0	projected national CRS	m	JSON
New Building	Building	Building	lod3MultiSurface		-	MultiSurface / CompositeSurface?	CityGML v.3	2 cm	LoD3.3/LoD1.X (sl	projected national CRS	m	JSON
New building - Façade	WallSurface	Building	semanticSurface?		-	MultiSurface / CompositeSurface?	CityGML v.3	2 cm	LoD2-3	projected national CRS	m	JSON
New building - Façade	WallSurface	Building	parent	to building	-	-	CityGML v.3	-	-	-	-	JSON
New building - Façade	WallSurface	Building Extension?	hasWindows		Boolean	-	CityGML v.3	-	-	-	-	JSON
New building - Roof	RoofSurface	Building	semanticSurface?		-	MultiSurface / CompositeSurface?	CityGML v.3	2 cm	LoD2-3	projected national CRS	m	JSON

Objective 2: Interoperabilità

City	Objects represented	Scale requested	Corresponding accuracy
APC	General positioning of the intervention on cartography	1:2000	0.4 m
APC	General context plan, including heights, plot, roads, public spaces, protected vegetation, surrounding buildings	1:500	0.1 m
APC	Plot plan and sections, including terrain	1:200	0.04 m
APC	Floor plans, sections and fronts	1:100	0.02 m
LIS	Site plan	1:200	0.04 m
LIS	Topographical survey	1:200	0.04 m
LIS	Architectural project	1:100 and 1:50	0.02 m and 0.01 m
GAI	Area of the building and surrounding	1:500 and 1:200	0.1 m and 0.04 m
GAI	Architectural project	1:100	0.02 m
PR	Wide Context plan	1:1000 - 1:50000	0.2 m – 10 m
PR	Cadastral situation	1:1000	0.2 m
PR	Coordination situation – general case	1:1000 – 1:200	0.2 m – 0.04 m
PR	Coordination situation – large buildings	1:5000 – 1:2000	1 m – 0.4 m
PR	Coordination situation – alteration to cultural heritage; conservation zone	1:200	0.04 m
PR	Architectural projects	1:100 and 1:50	0.02 m and 0.01 m

Geometry Accuracy
2 cm
3 10 cm
3 -
?
?
3 10 cm
3 -
-
3 10 cm
3 4 cm
4 cm
3 4 cm
3 10 cm
3 2 cm
3 2 cm
3 -
-
3 2 cm

Spatial resolutio
LoD0
LoD0
-
-
-
-
-
-
LoD0



CityGML

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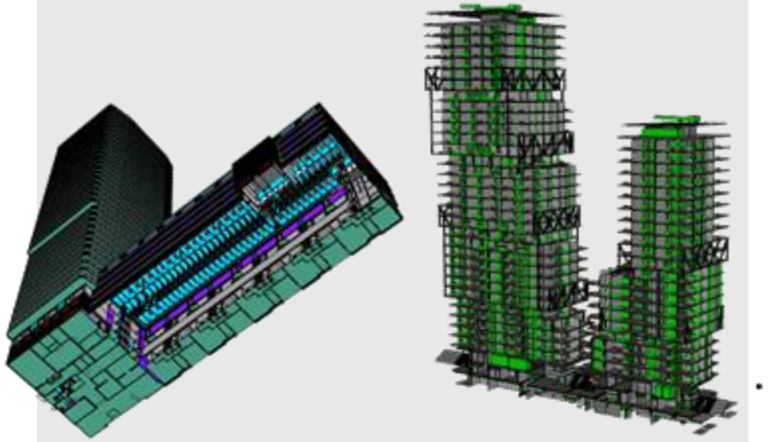
Data about the building:

Building height
Rooms volumes
Windows dimensions
Escape routes
...

Building Information Models



Data about the context:

Roads attributes
Public places accessibility
Public transport information
Protected areas
Noise sources
...

(3D) Geographical Information Systems

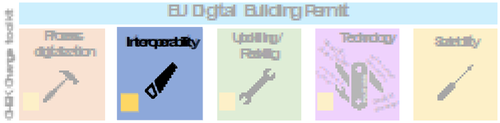
Regulations:

Constraints to be respected and checked for building permit issuing

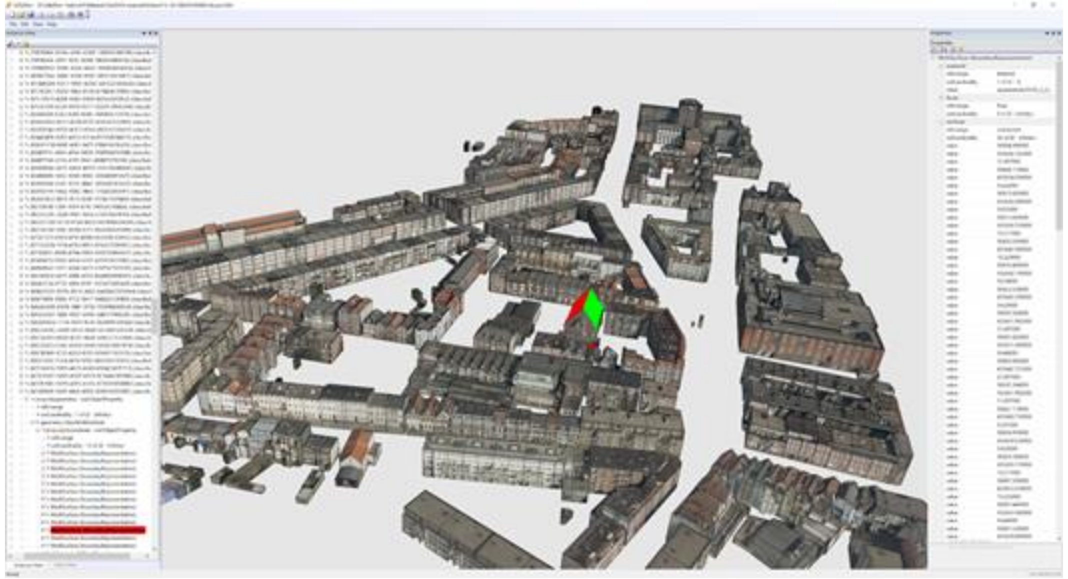
Formal machine-readable (spatialized) regulations

Integration / Interoperability

Objective 2: Interoperability



Geoinformation ←→ BIM



Objective 3: Formazione dei diversi attori nel caso d’uso

CHEK Project - eLearning Hub

Available courses

Municipality Technicians

Explore the future of municipal permitting with our Chek Toolkit Training for Municipality Technicians. This dynamic course is not just an exploration but a roadmap to the future, offering participants a comprehensive understanding of digital building permits and their pivotal role in modernizing municipal processes. With a focus on essential technologies such as BIM and GIS, participants will gain insight about the benefits, challenges, and methods of digitizing permit rules through real-world examples and hands-on exercises. By the end of the course, participants will not only gain skills, but also insights and methods for navigating the complexities of digital building permits with confidence and improved outcomes in their municipalities.

Designers (Applicants)

Through our Chek Toolkit Training, you'll be able to engage in an exploration into the world of digital permitting that is specifically suited to designers. Throughout this training, participants will delve into the intricacies of digital building permits, with an emphasis on essential technologies such as BIM and GIS that are vital for designers. This course provides an in-depth examination of the benefits, challenges, and methodologies involved in digitizing permit protocols. Through a blend of theoretical discussions, practical demonstrations, and hands-on exercises, participants will gain practical insights into navigating the complexities of digital permitting systems. Real-world examples and case studies will be utilized to illustrate key concepts and best practices. By the completion of the course, designers will possess the necessary abilities and expertise needed to effectively deal with digital building permits.

Other stakeholders in the construction value chain

Explore the transformational potential of digital building permits with our Chek Toolkit Training, which is targeted for construction value chain stakeholders. Learn about the benefits of digital permits through worldwide examples. Join us in leveraging technology to accelerate the construction permits process and improve efficiency and transparency in the construction

×

Registration For Advanced Summer Course Is Open

Digital Transformation in Building Permits: Advanced Practices and the CHEK Framework is an intensive one-week **(07-11 July 2025, GUIMARÃES, Portugal)** summer course designed to equip participants with cutting-edge knowledge and practical skills in the digitalization of building permit process

Learn More

https://elearning.chekdbp.eu

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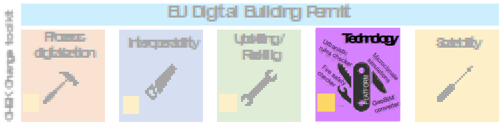
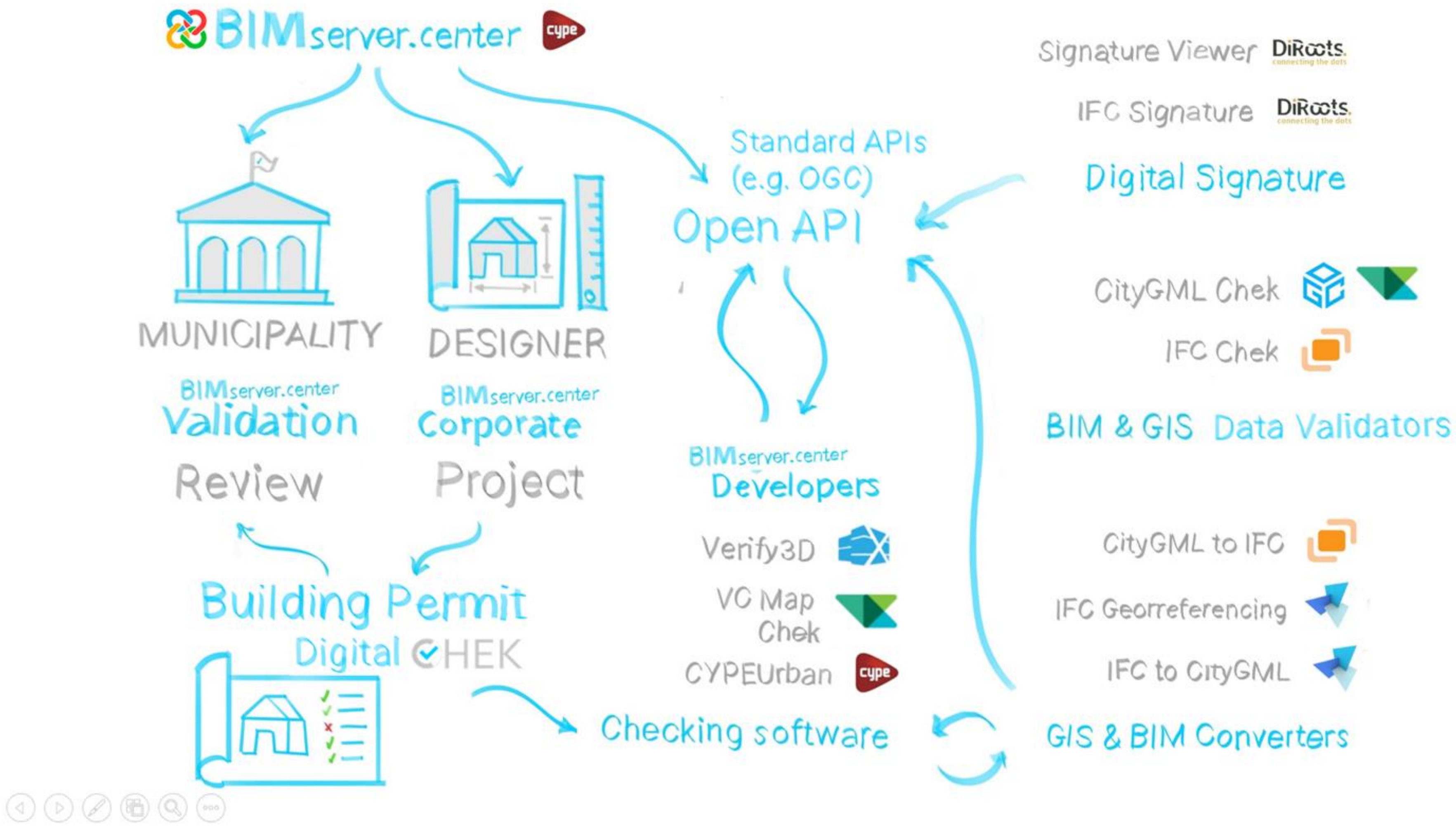
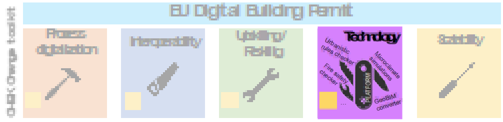
Objective 4: Integrated software tools

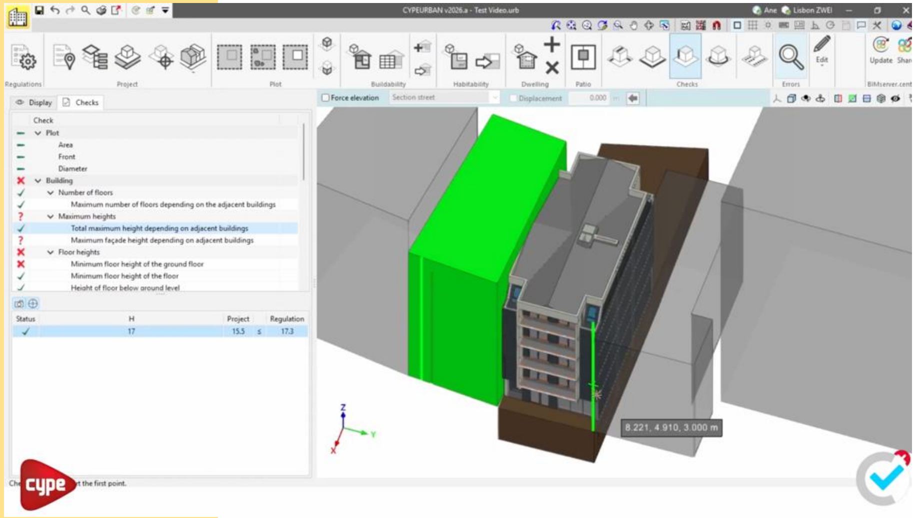


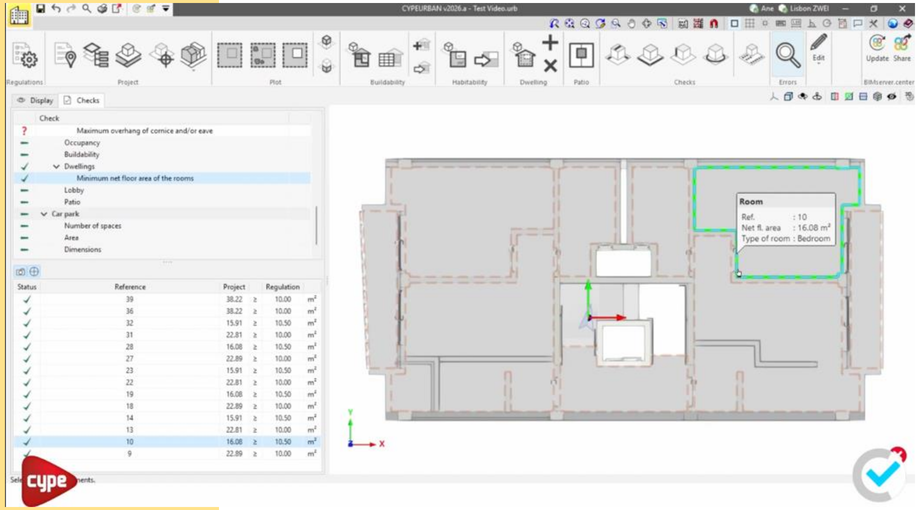


CHEK DIGITAL BUILDING PERMIT
https://chekdbp.eu

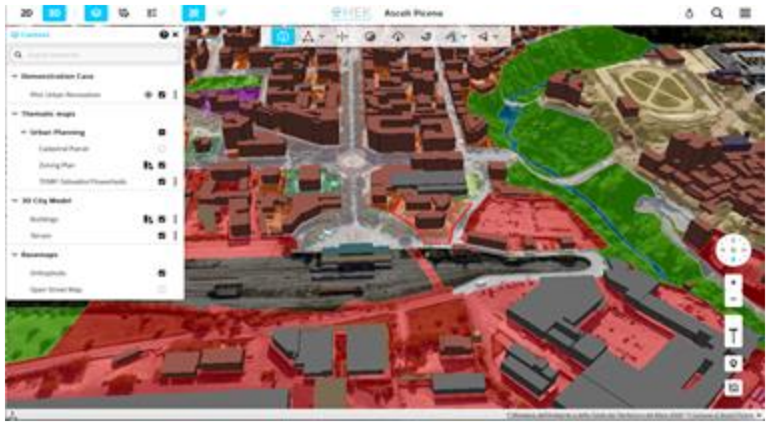
Objective 4: Integrated software tools

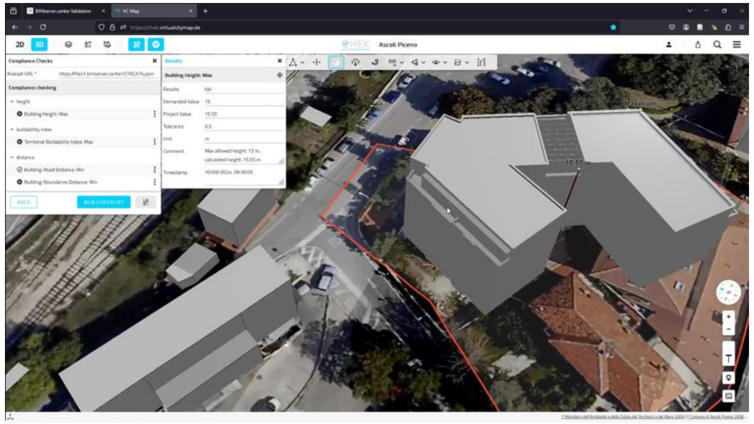


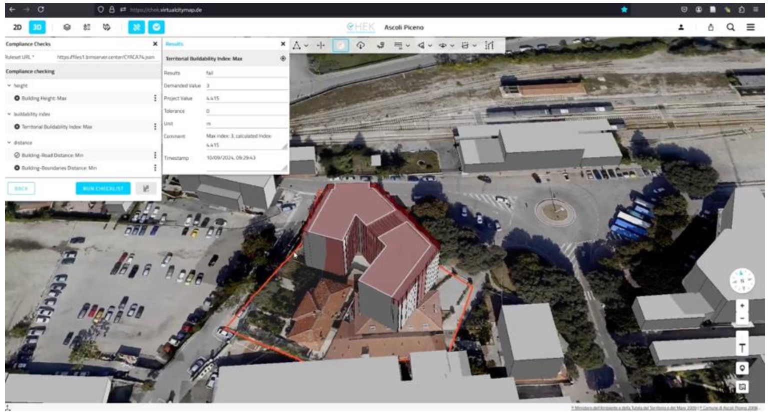


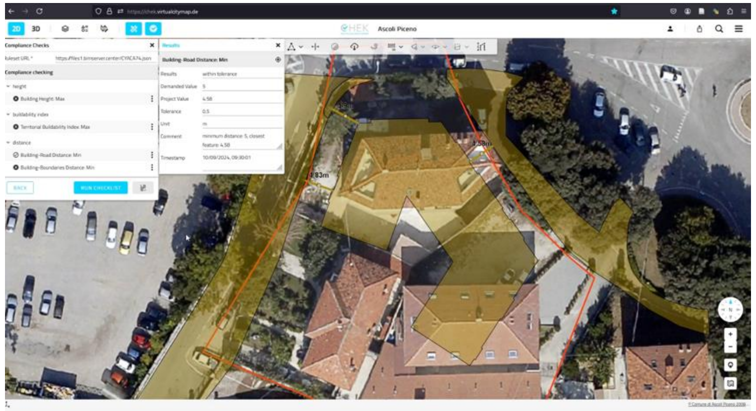


virtual city systems









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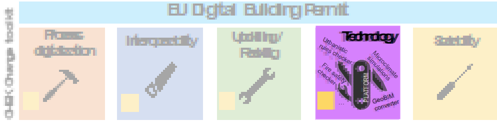
Objective 4: Integrated software tools



Abilitare **analisi complesse**, attualmente difficilmente verificabili oggettivamente: parametri urbanistici dipendenti da un contesto territoriale più ampio, ombre, viste, parametri ambientali.

Application	Status of the application	Visibility	Version	GUID
CYPELEC Core	Authorised application		2024.e	
CYPE Memorias CTE	Authorised application		2024.e	
CYPESOUND CTE	Authorised application		2024.e	
StruBIM Embedded Walls	Authorised application		2024.e	
StruBIM Foundations	Authorised application		2024.e	
CYPESOUND NRA	Authorised application		2024.e	
CYPEURBAN	Authorised application		2024.e	
CYPELEC Grounding IEC	Authorised application		2024.e	

Objective 5: Scalability

Dimostrazioni e scalabilità

CHEK community of practice:

https://chekdbp.eu/?page_id=5734

Lisbon (PT)

Vila Nova de Gaia (PT)

Ascoli Piceno (IT)

Prague (CZ)

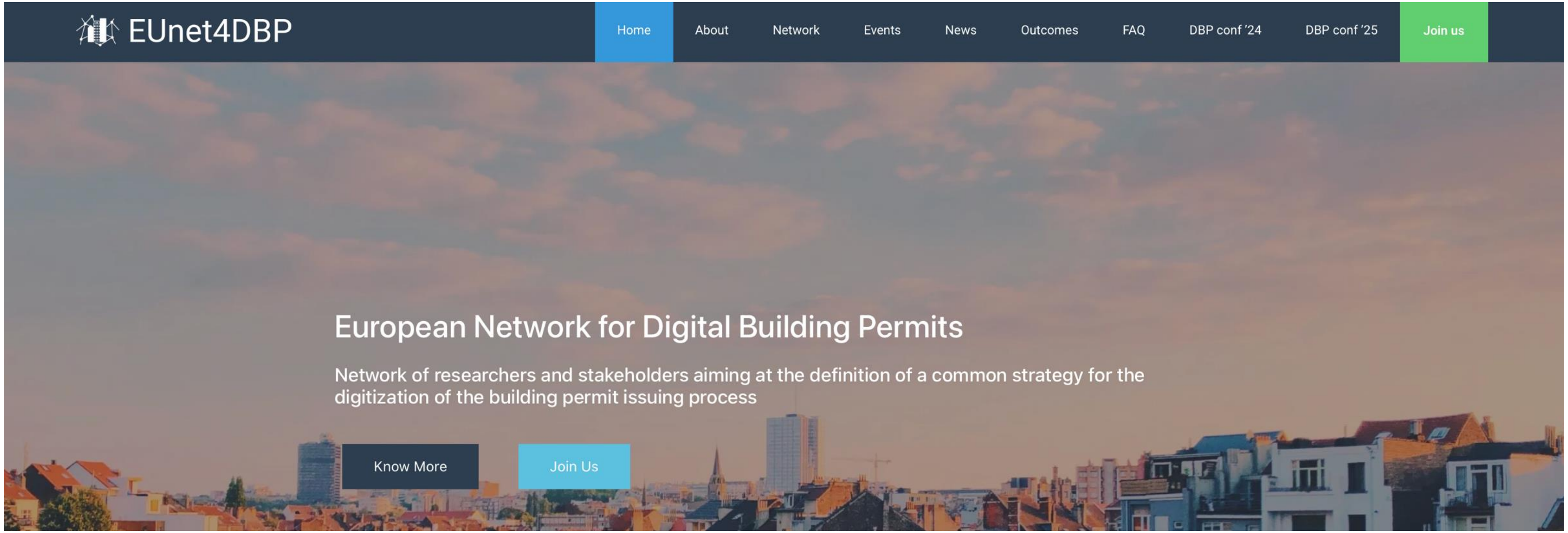
This project has received funding from the European Union's Horizon Europe programme under Grant Agreement No.101058559

Funded by the European Union

European Network for Digital Building Permit - EUnet4DBP

EUnet4DBP

https://eu4dbp.net



Start 2020

Mission - To **accelerate the digital transformation** of the building permit process.

Vision - Development of **flexible, scalable and re-usable** digital and (semi-) automated building permit tools and methods in a common effort and under the umbrella of an open science framework, by sharing experiences and building-up knowledge.

Thank You

Community

500+ International Members

110+ Member Meetings

60+ Alliance and Liaison partners

50+ Standards Working Groups

45+ Domain Working Groups

25+ Years of Not for Profit Work

10+ Regional and Country Forums

Innovation

120+ Innovation Initiatives

380+ Technical reports

Quarterly Tech Trends monitoring

Standards

65+ Adopted Standards

300+ products with 1000+ certified implementations

1,700,000+ Operational Data Sets

Using OGC Standards

https://www.ogc.org